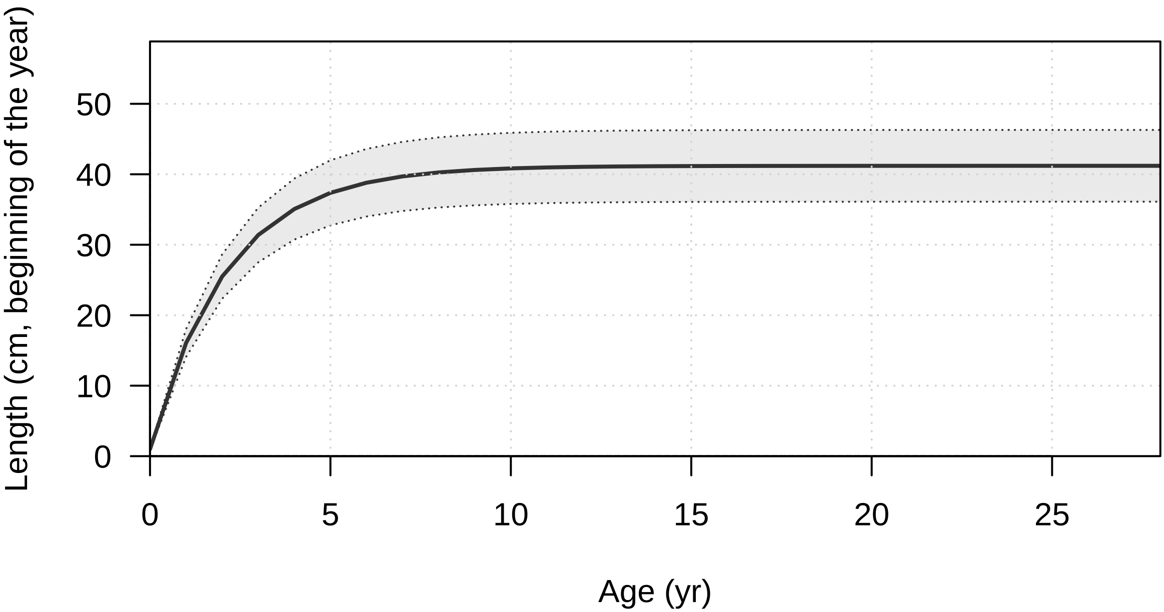
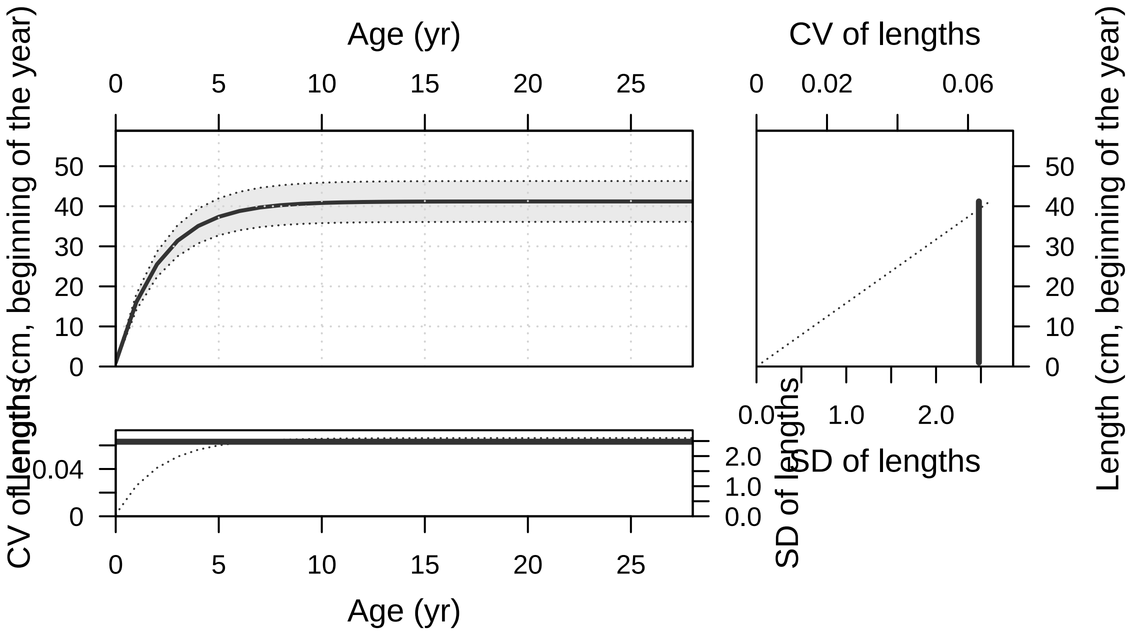
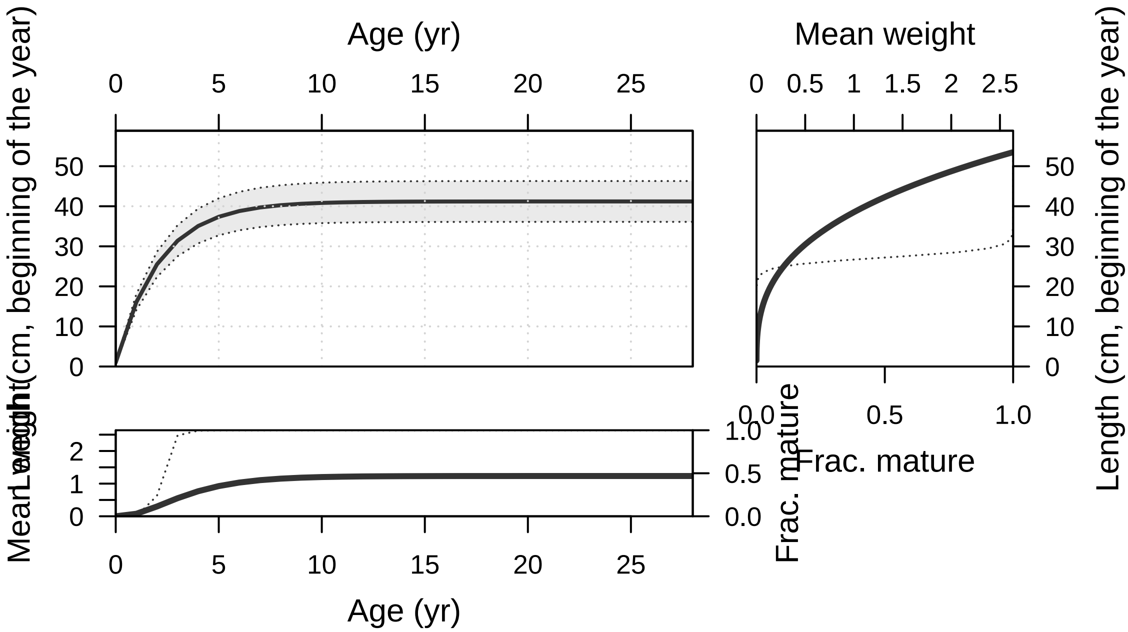
Plots created using the 'r4ss' package in R
Stock Synthesis version: 3.30.19.0
StartTime: Fri Feb 13 12:03:00 2026
Data_File: data.ss
Control_File: control.ss



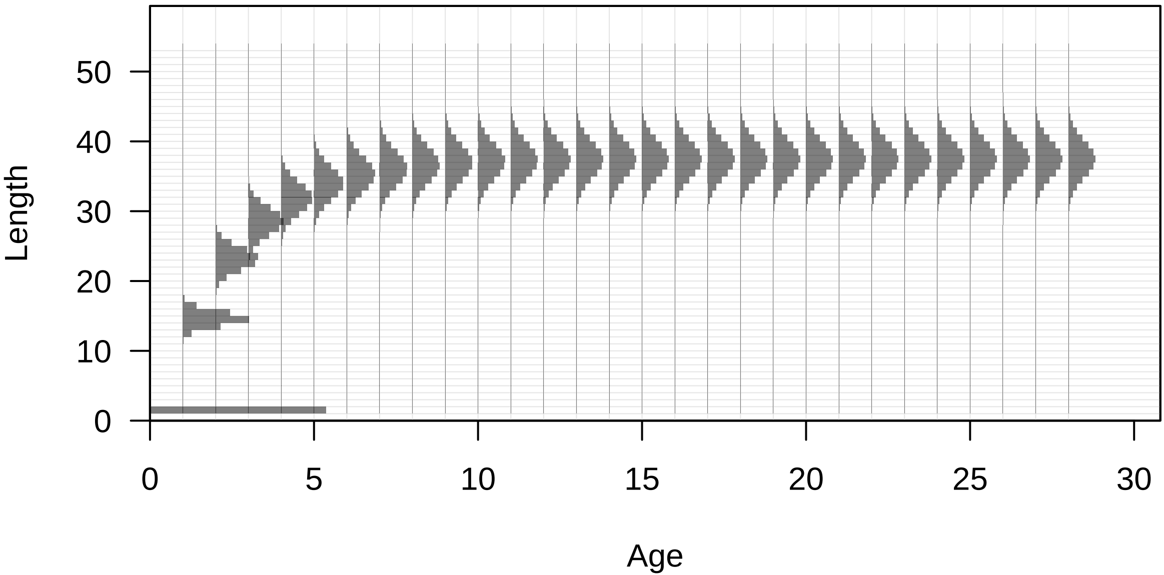
Length (cm, beginning of the year)
0
10
20
30
40
50
0
5
10
15
20
25
Age (yr)



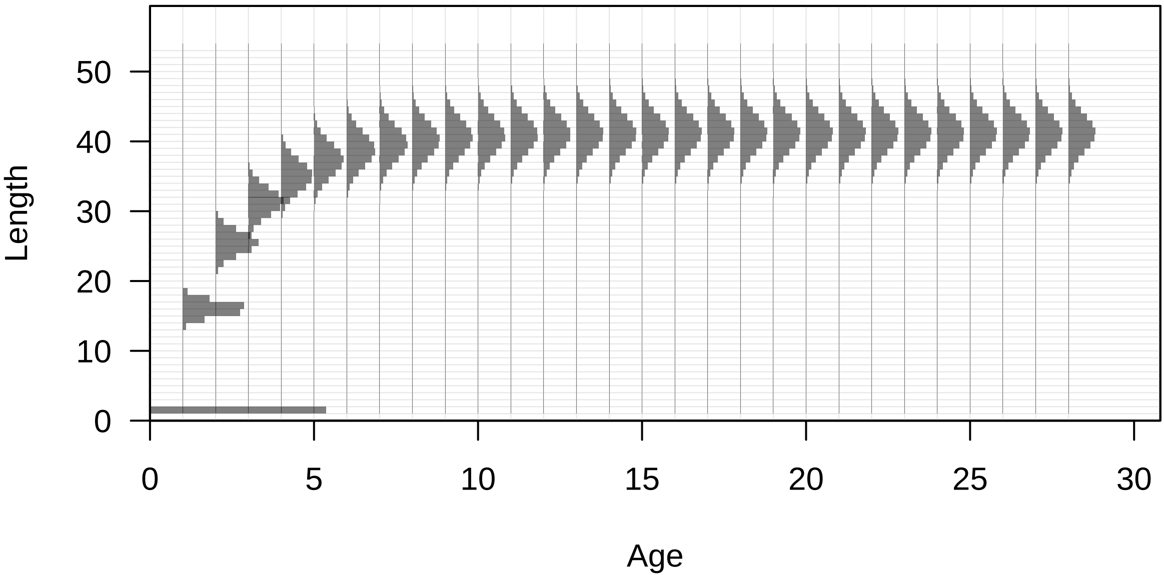
Age (yr)
0
5
10
15
20
25
CV of lengths
0
0.02
0.06
Length (cm, beginning of the year)
50
40
30
20
10
0
CV of lengths
0.04
0
SD of lengths
2.0
1.0
0.0
Age (yr)
SD of lengths
0.0
1.0
2.0
SD of lengths
Length (cm, beginning of the year)



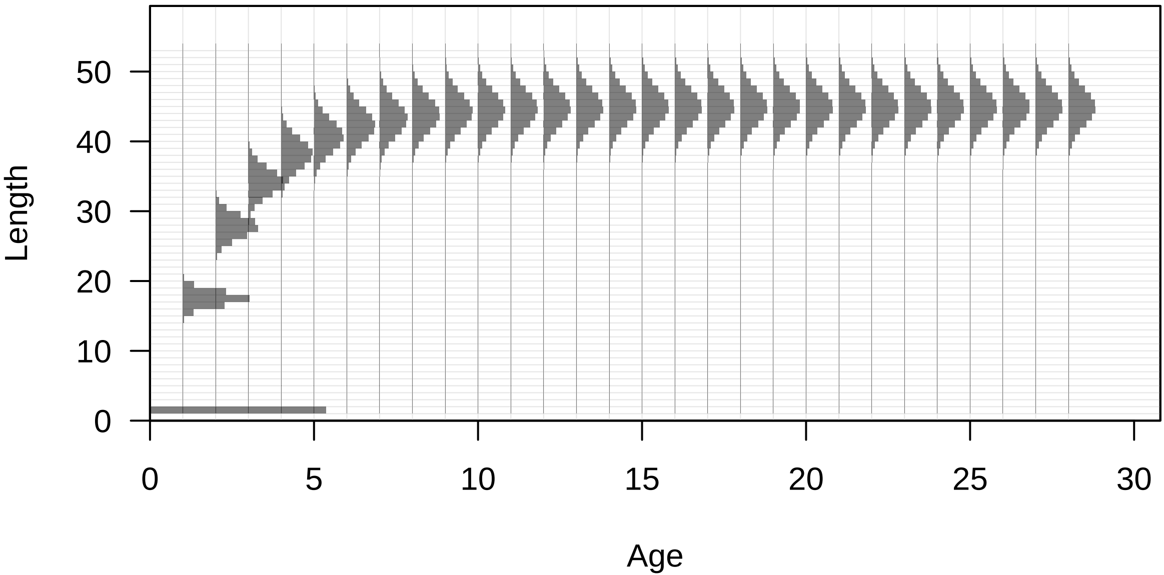
Age (yr)
Mean weight
0
5
10
15
20
25
0
0.5
1
1.5
2
2.5
Length (cm, beginning of the year)
50
40
30
20
10
0
0.0
0.5
1.0
Frac. mature
Mean weight
Length
2
1
0
1.0
0.5
0.0
Frac. mature
Age (yr)



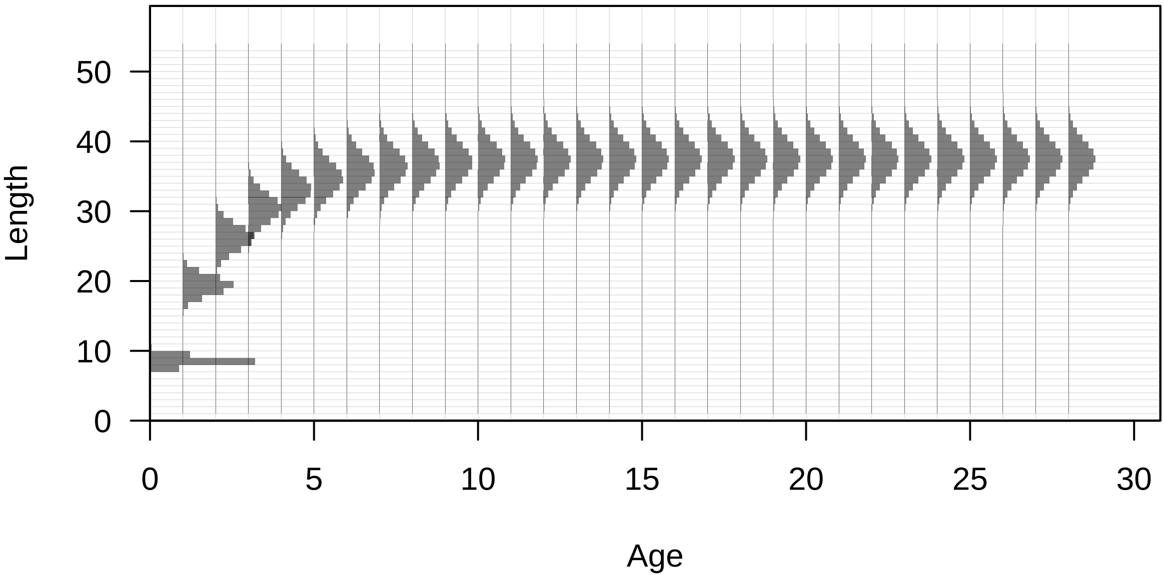
Length
Age
0
10
20
30
40
50
0
5
10
15
20
25
30



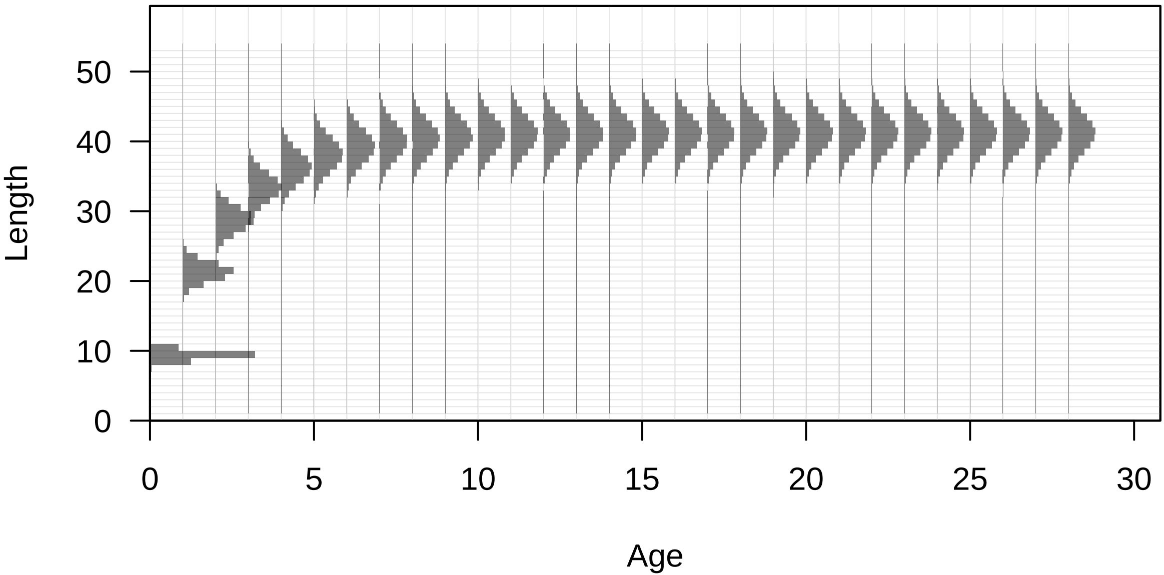
Length
50
40
30
20
10
0
0
5
10
15
20
25
30
Age



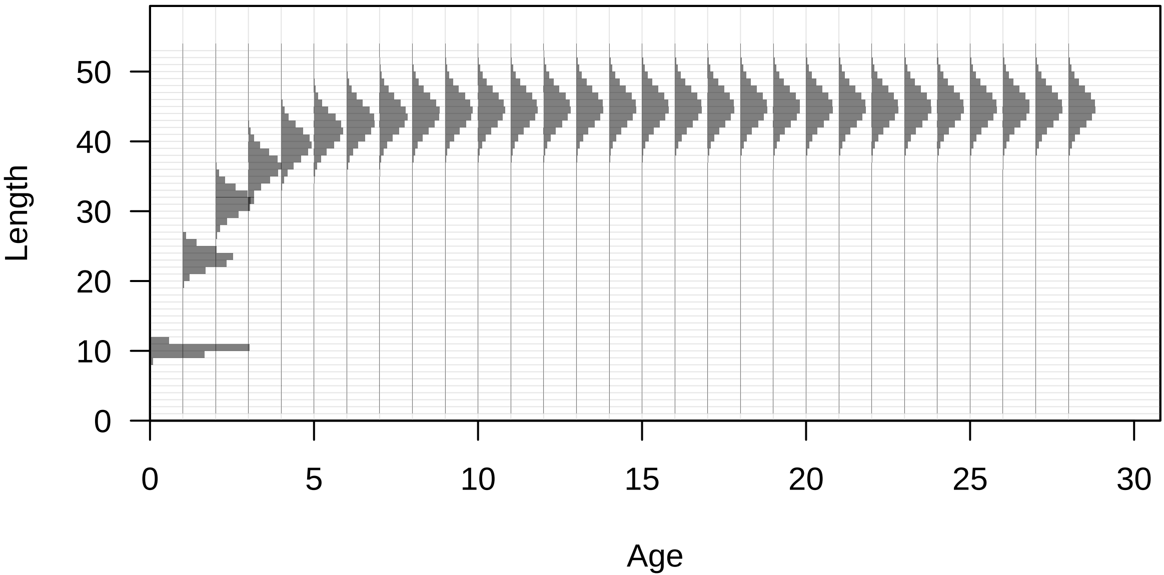
Length
50
40
30
20
10
0
0
5
10
15
20
25
30
Age



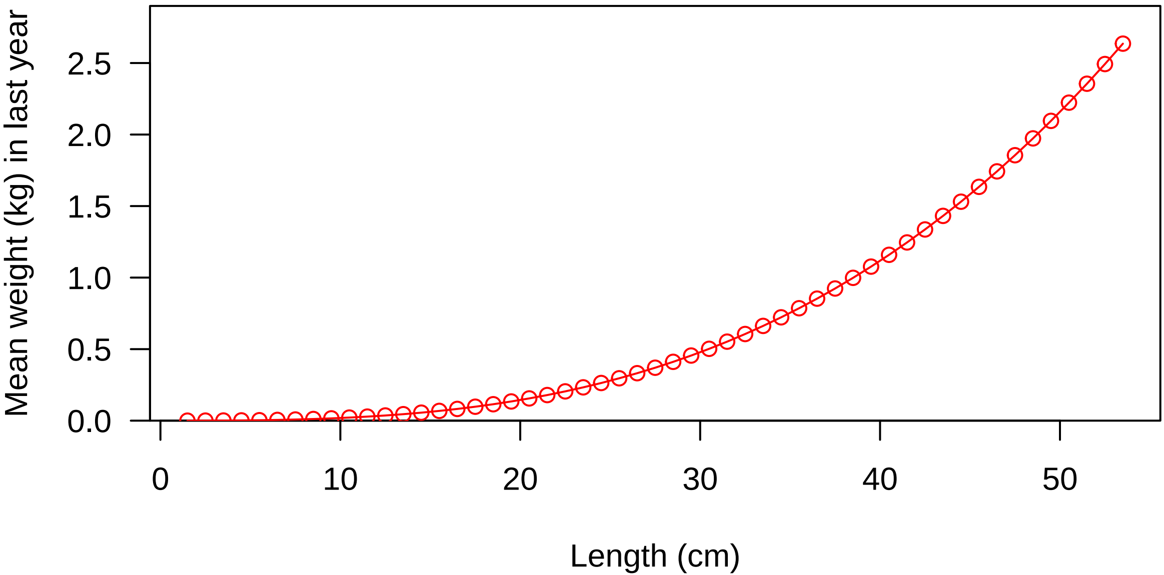
Length
0
10
20
30
40
50
Age
0
5
10
15
20
25
30

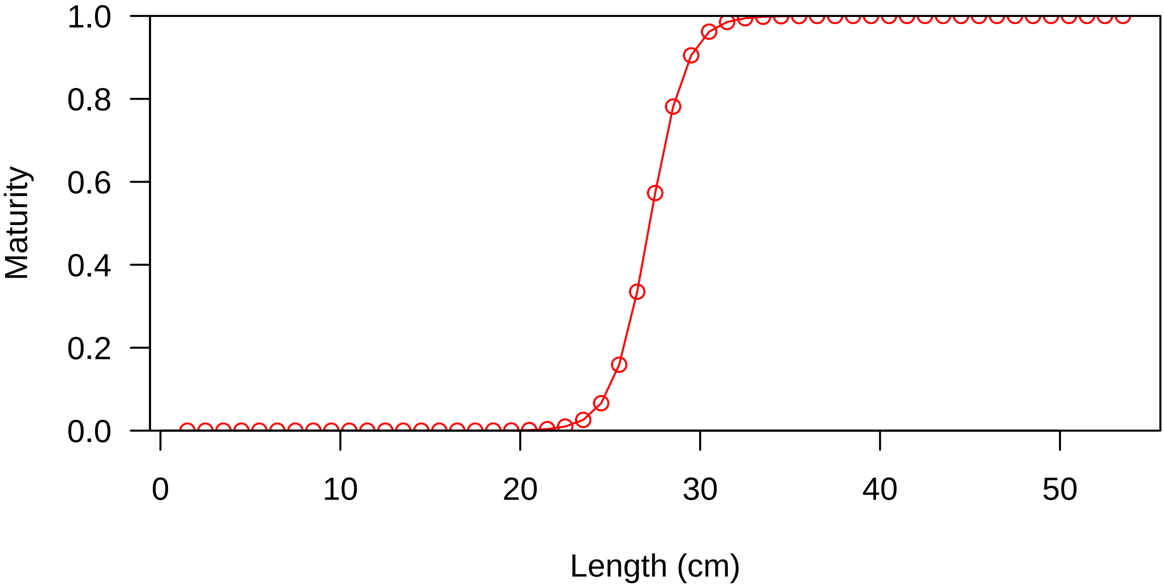


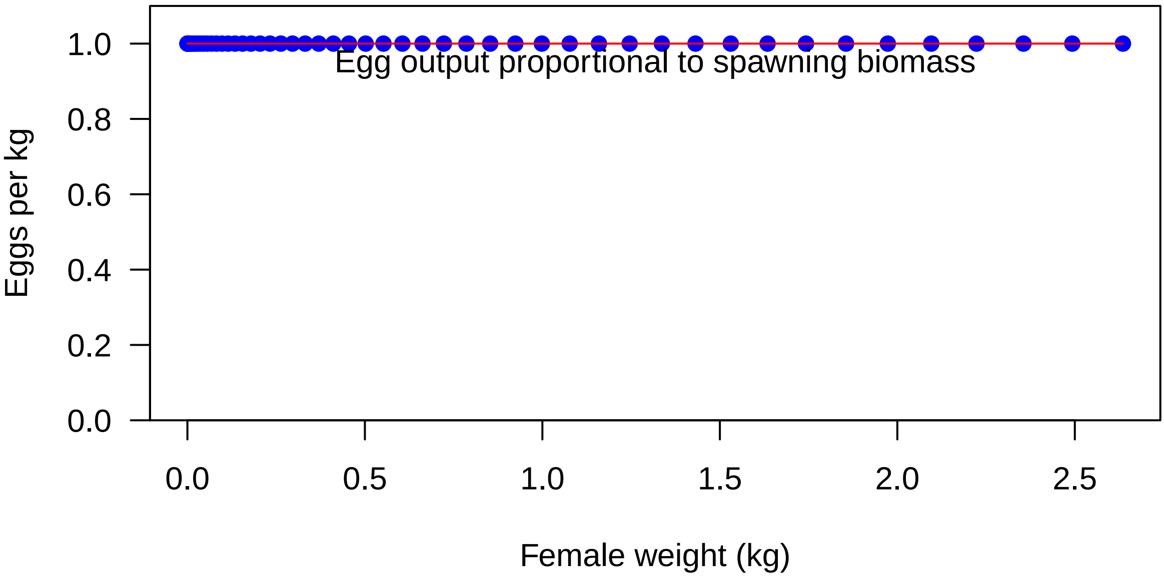
Length
50
40
30
20
10
0
0
5
10
15
20
25
30
Age



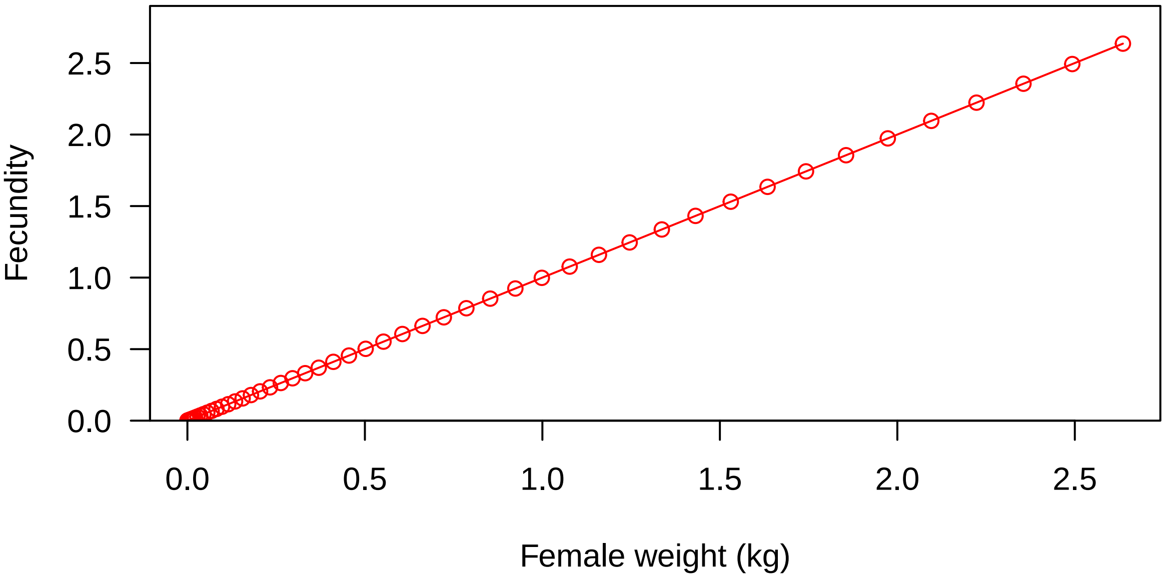
Length
Age
0
10
20
30
40
50
0
5
10
15
20
25
30


Mean weight (kg) in last year
0.0
0.5
1.0
1.5
2.0
2.5
0
10
20
30
40
50
Length (cm)

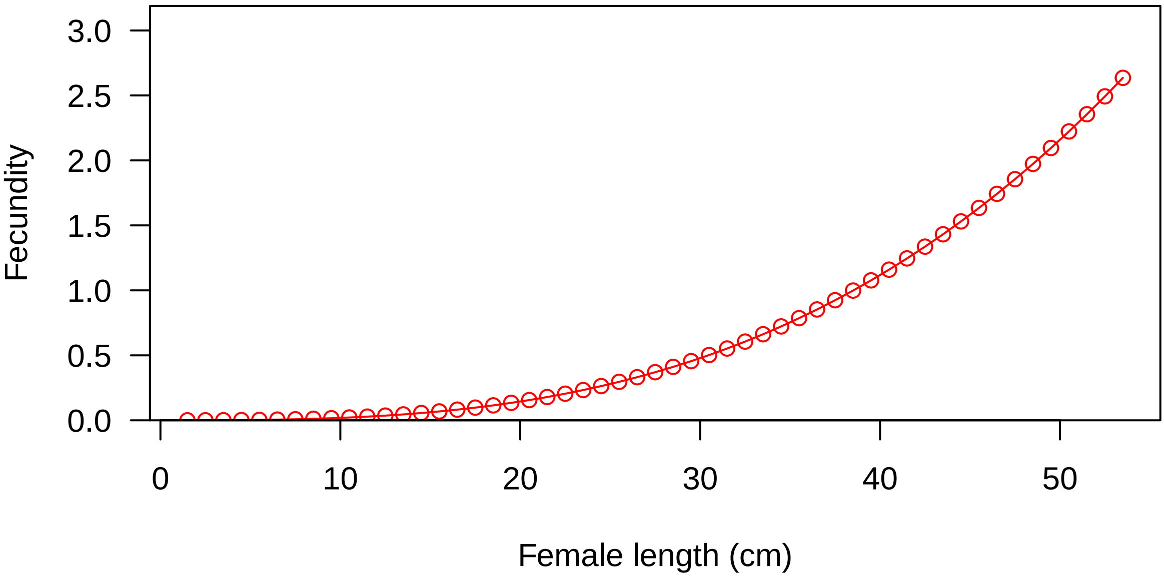

1.0
0.8
0.6
0.4
0.2
0.0
Maturity
0
10
20
30
40
50
Length (cm)



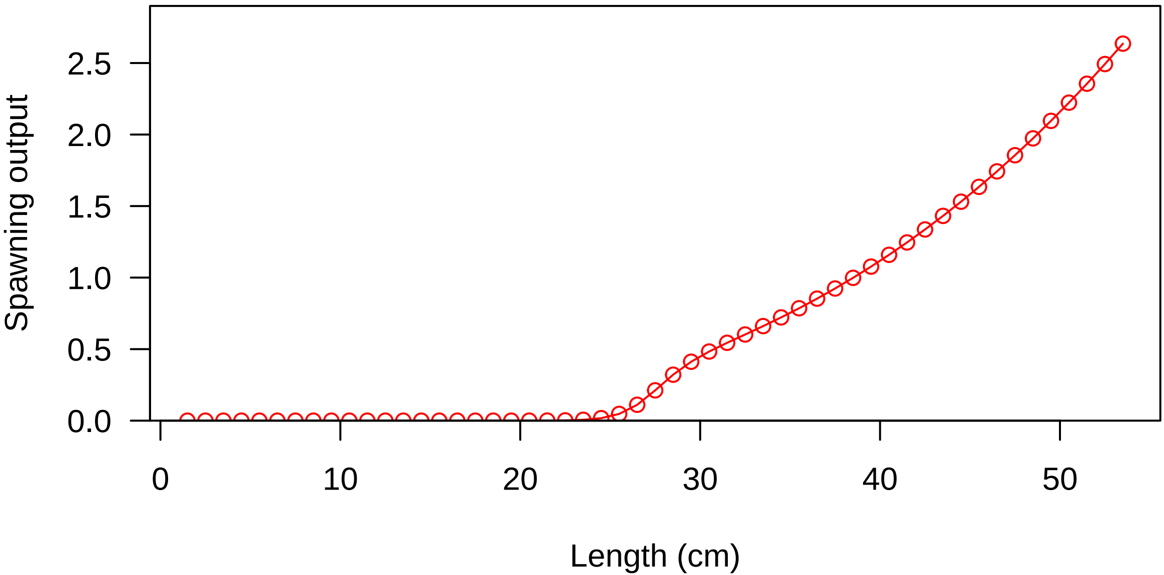
Egg output proportional to spawning biomass
Eggs per kg
Female weight (kg)
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5
2.0
2.5



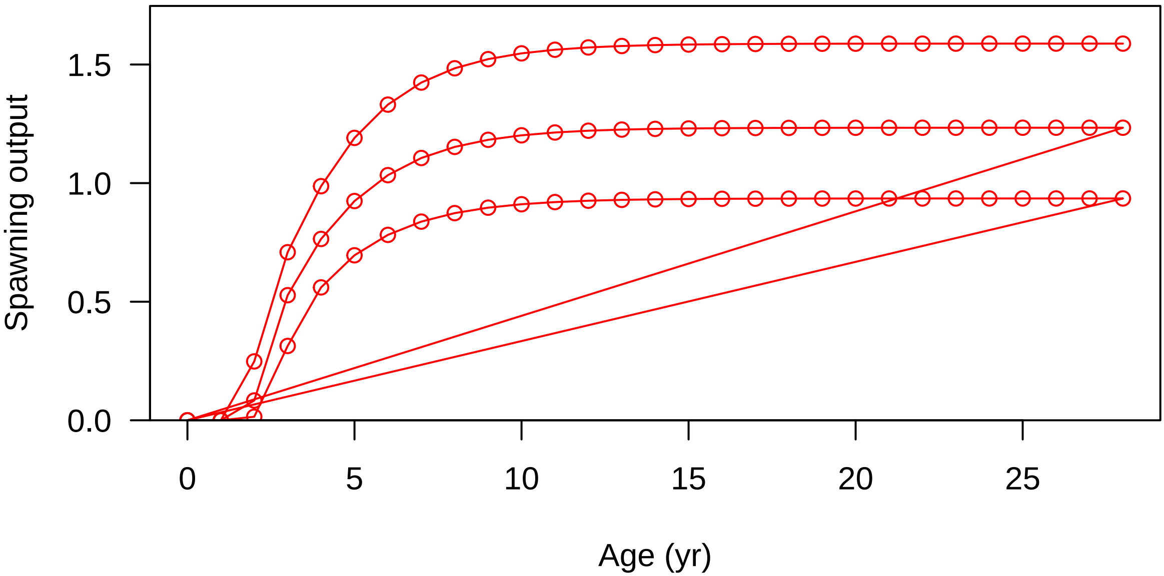
Fecundity
2.5
2.0
1.5
1.0
0.5
0.0
0.0
0.5
1.0
1.5
2.0
2.5
Female weight (kg)



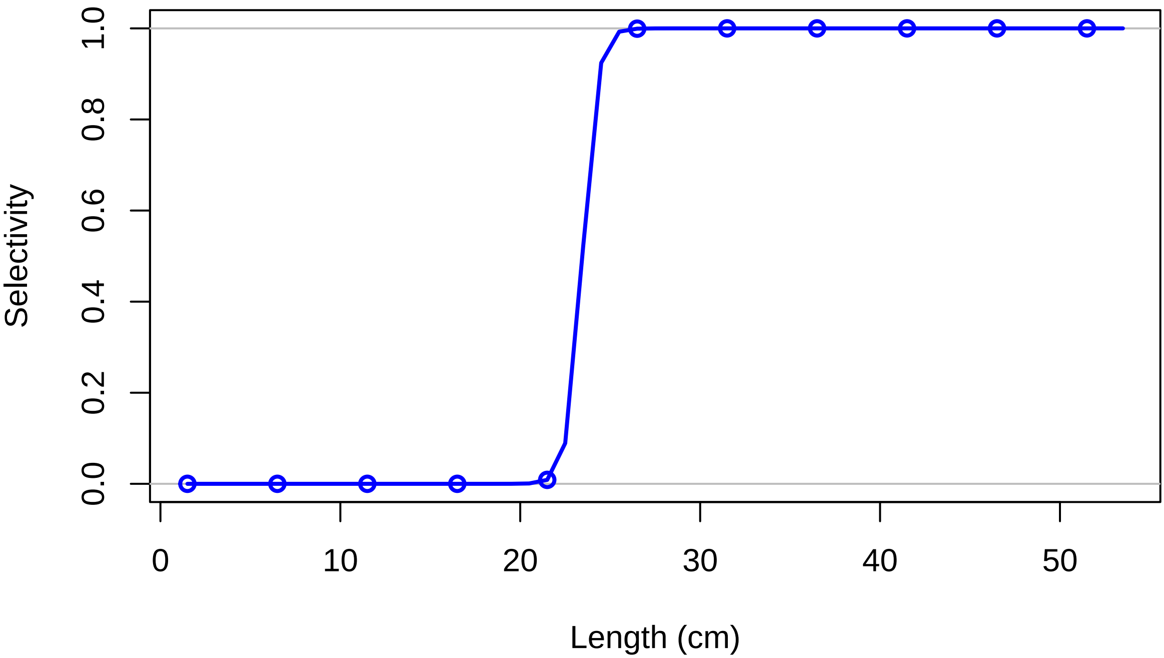
Fecundity
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
10
20
30
40
50
Female length (cm)



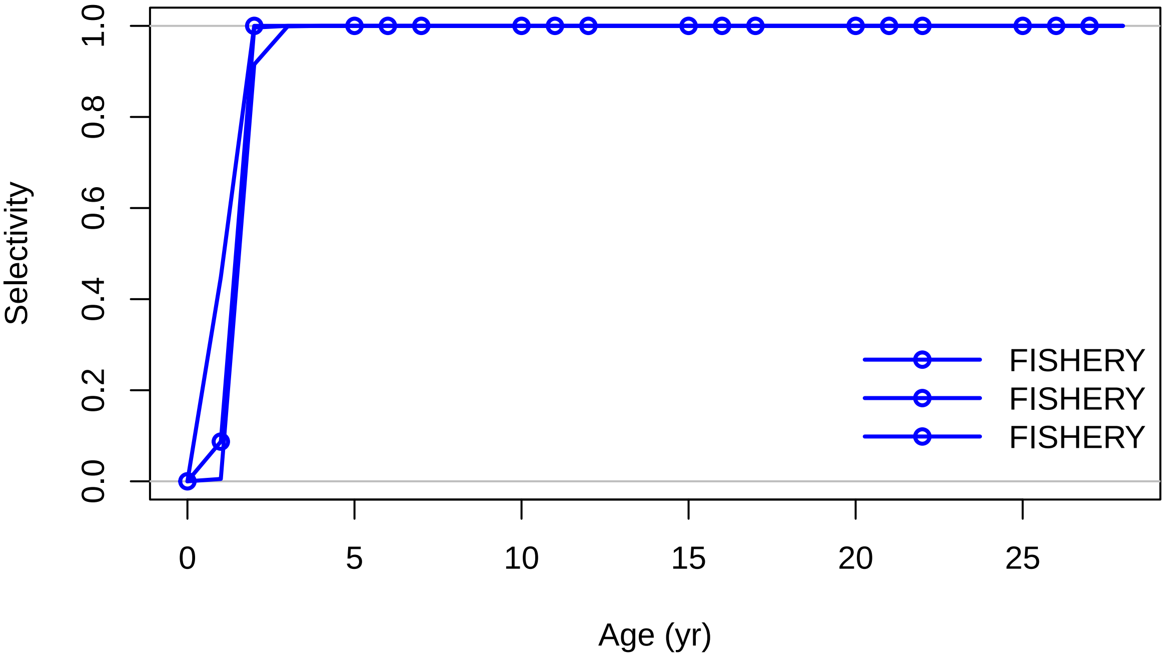
Spawning output
0.0
0.5
1.0
1.5
2.0
2.5
0
10
20
30
40
50
Length (cm)

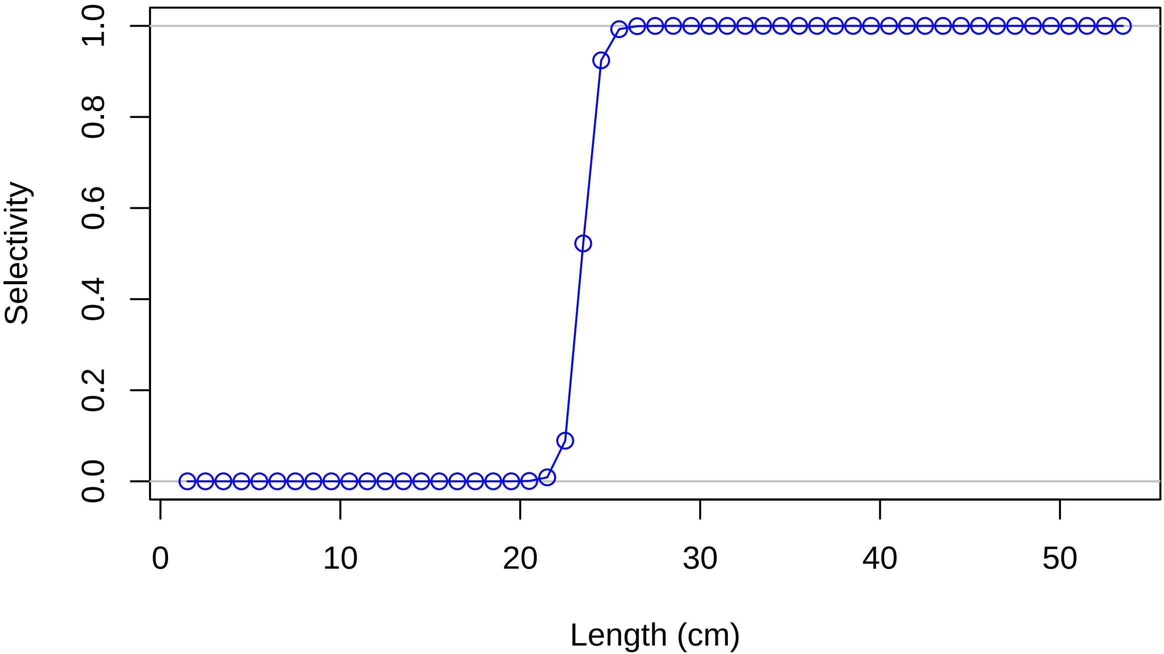


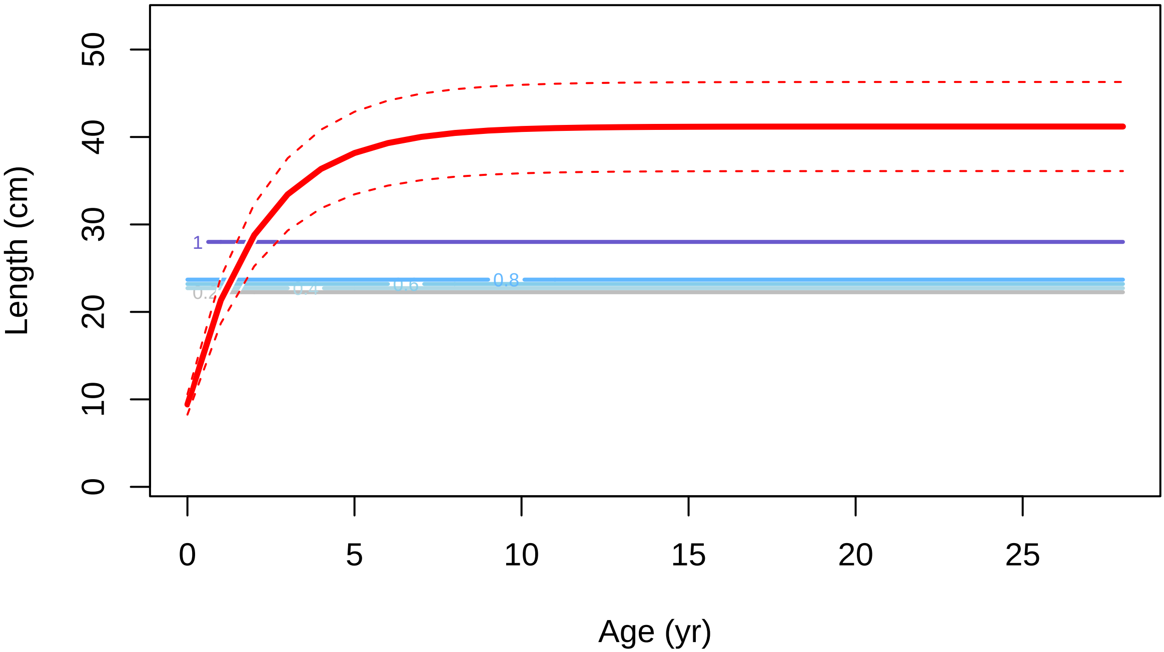
Spawning output
1.5
1.0
0.5
0.0
0
5
10
15
20
25
Age (yr)



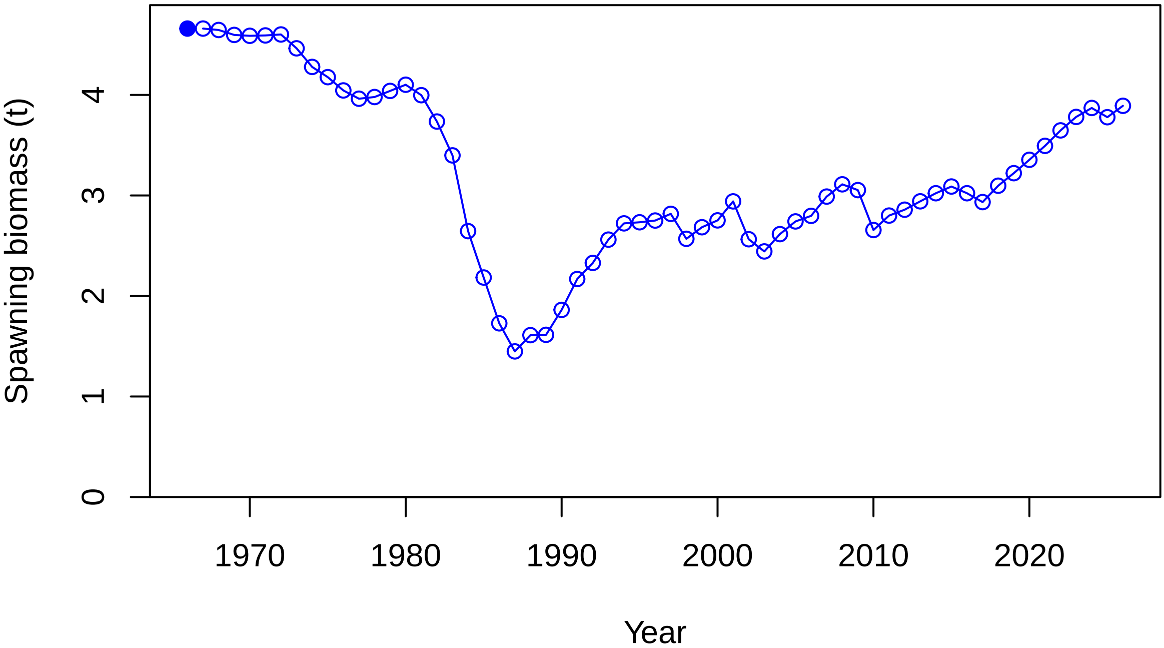
Selectivity
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
50
Length (cm)


Selectivity
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25
Age (yr)
FISHERY
FISHERY
FISHERY

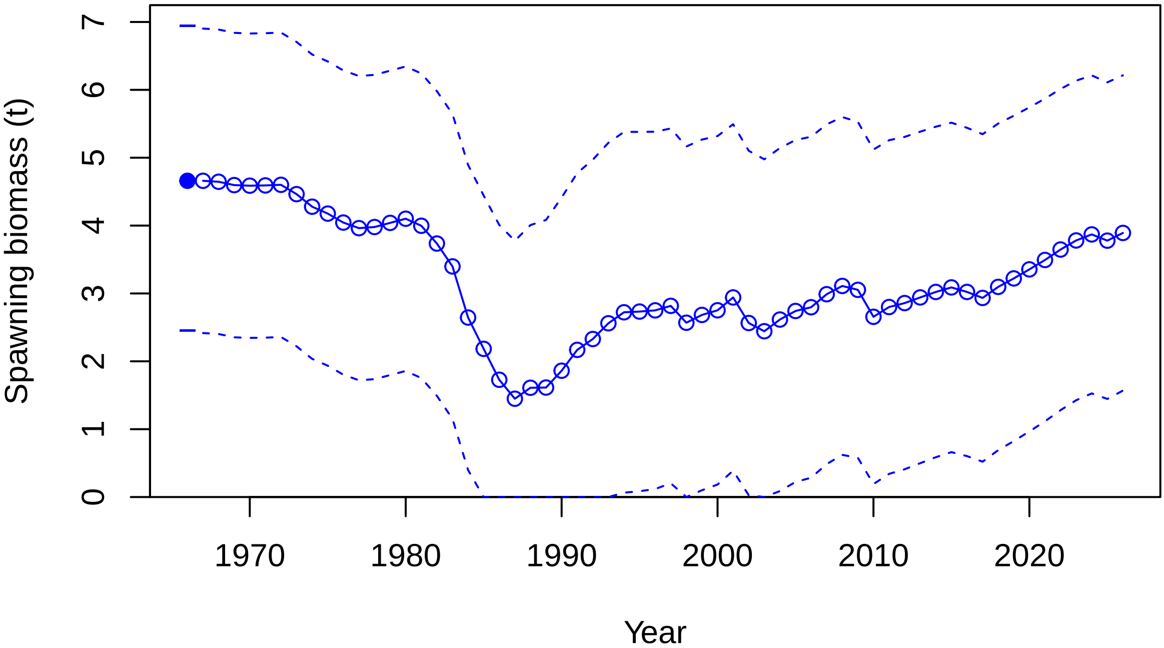

Selectivity
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
50
Length (cm)



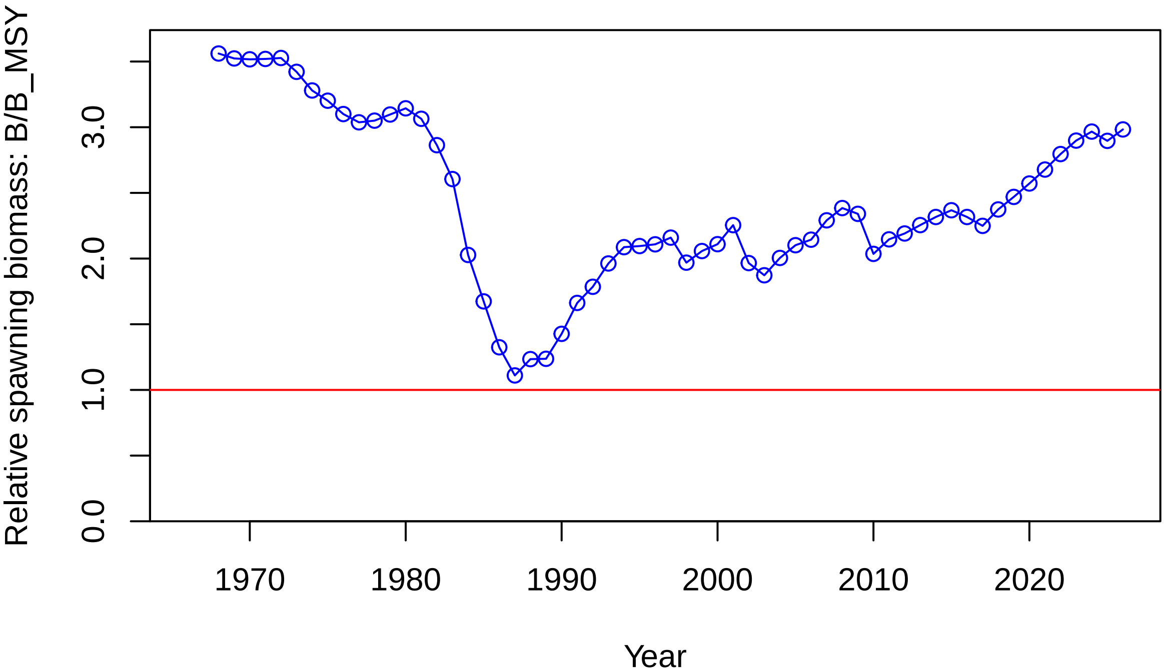
Length (cm)
Age (yr)
0
10
20
30
40
50
0
5
10
15
20
25
1
0.8
0.6
0.4
0.2



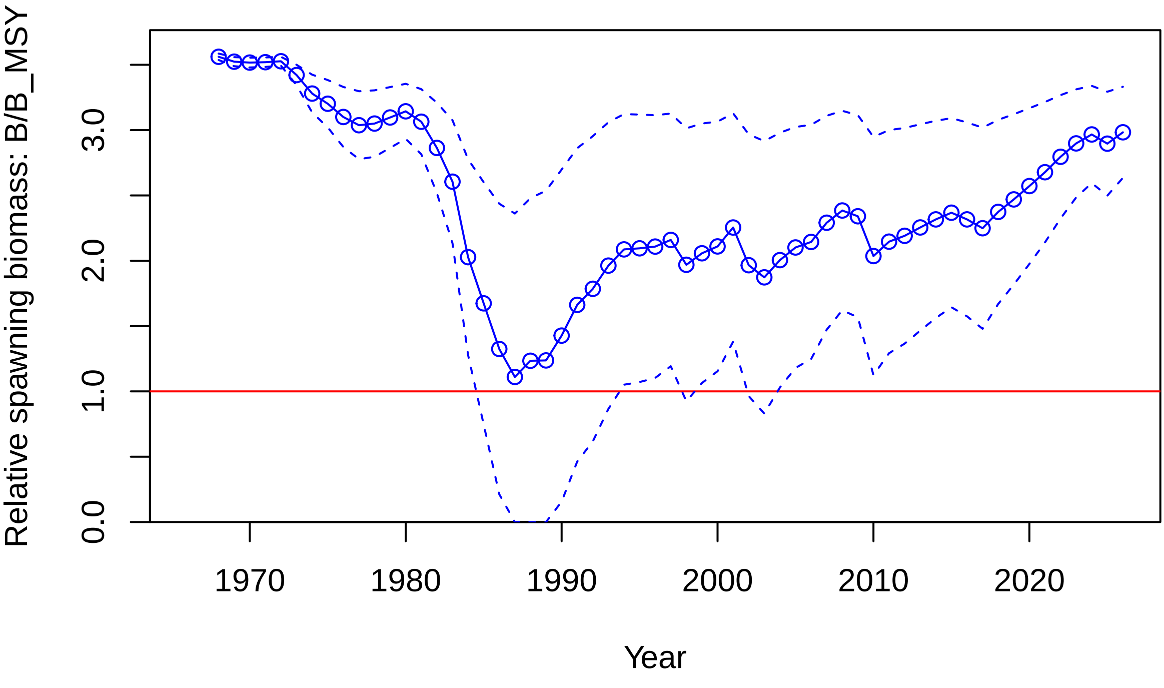
Spawning biomass (t)
0
1
2
3
4
1970
1980
1990
2000
2010
2020
Year



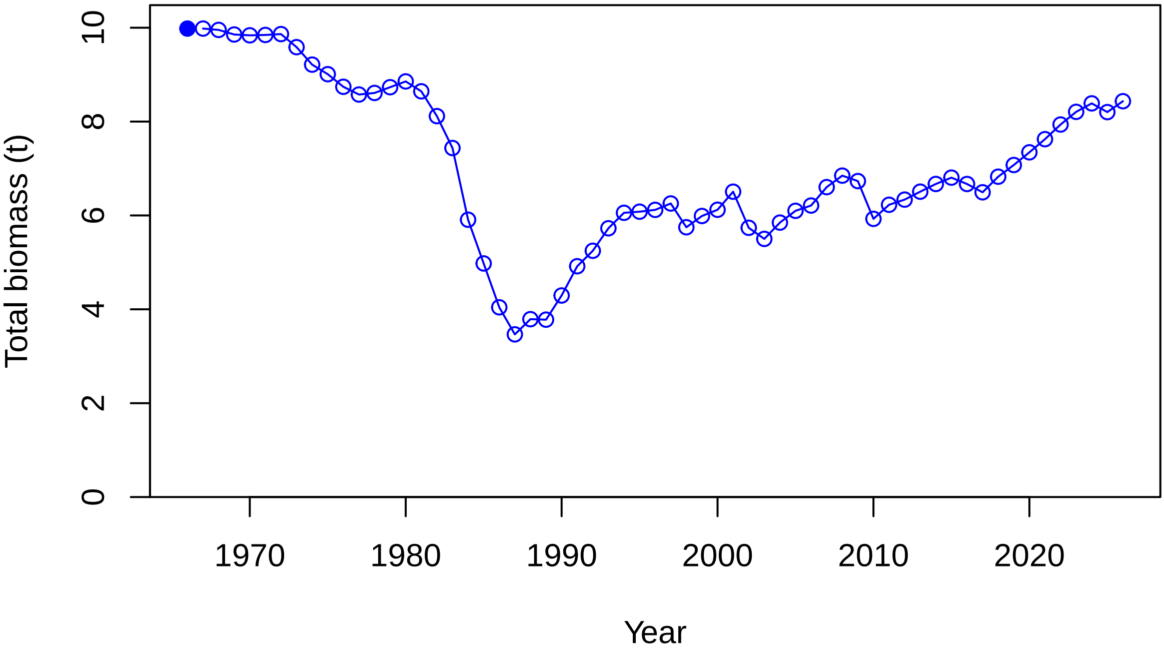
Spawning biomass (t)
0
1
2
3
4
5
6
7
1970
1980
1990
2000
2010
2020
Year

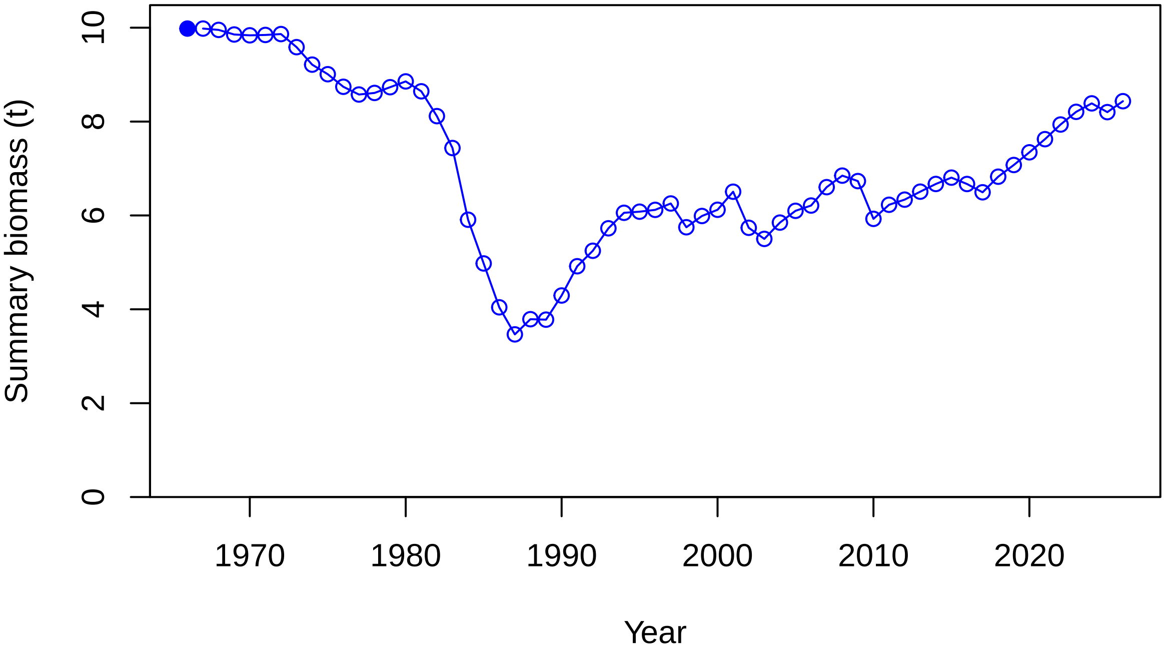


Relative spawning biomass: B/B_MSY
0.0
1.0
2.0
3.0
1970
1980
1990
2000
2010
2020
Year

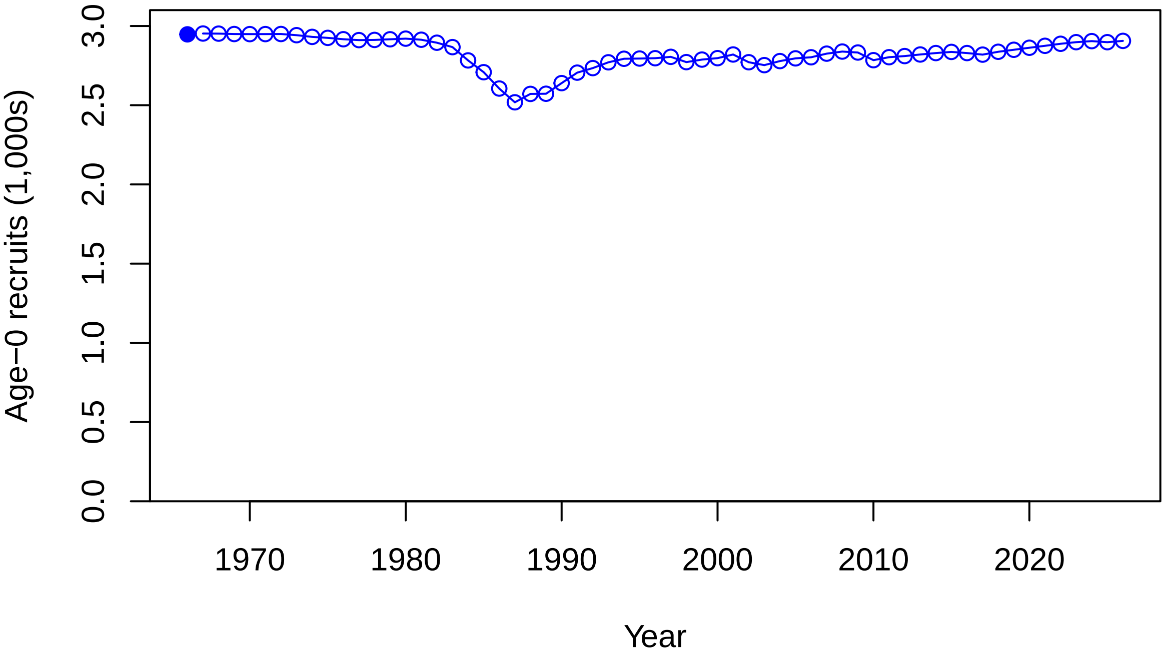


Relative spawning biomass: B/B_MSY
0.0
1.0
2.0
3.0
1970
1980
1990
2000
2010
2020
Year

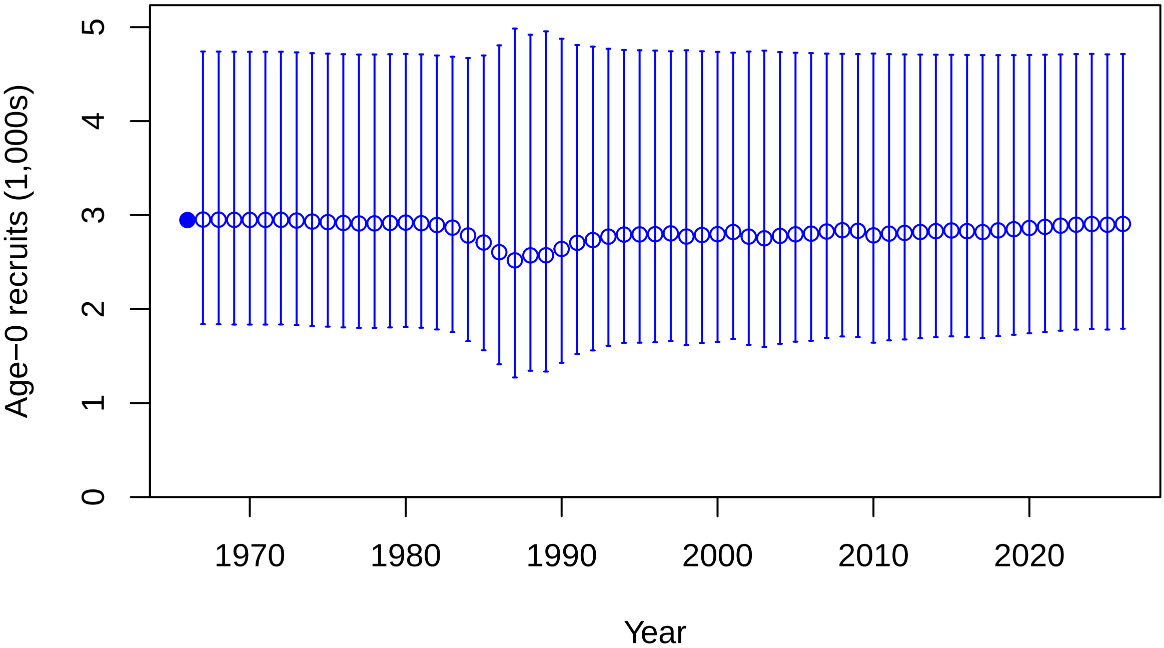

Total biomass (t)
0
2
4
6
8
10
1970
1980
1990
2000
2010
2020
Year



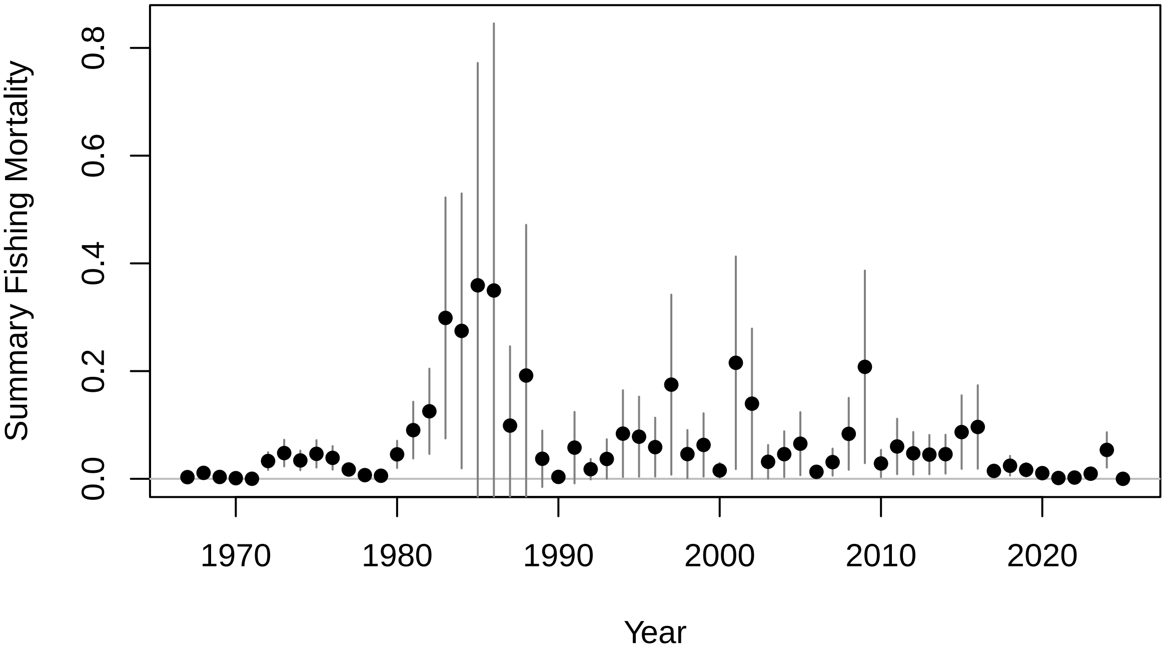
Summary biomass (t)
0
2
4
6
8
10
1970
1980
1990
2000
2010
2020
Year

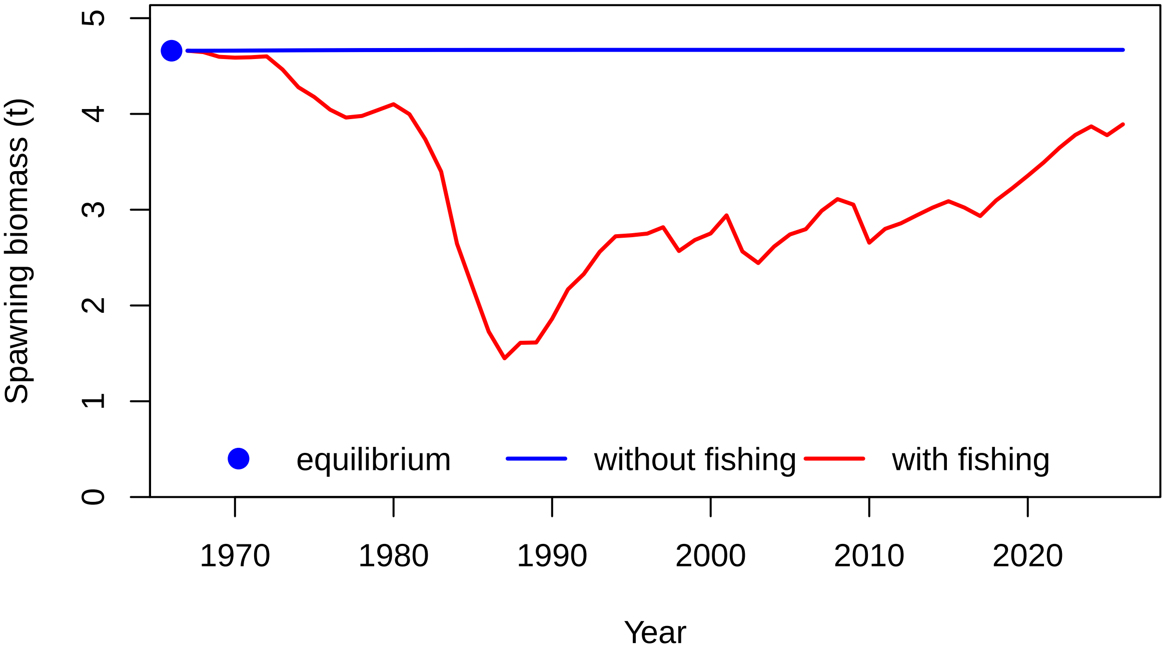


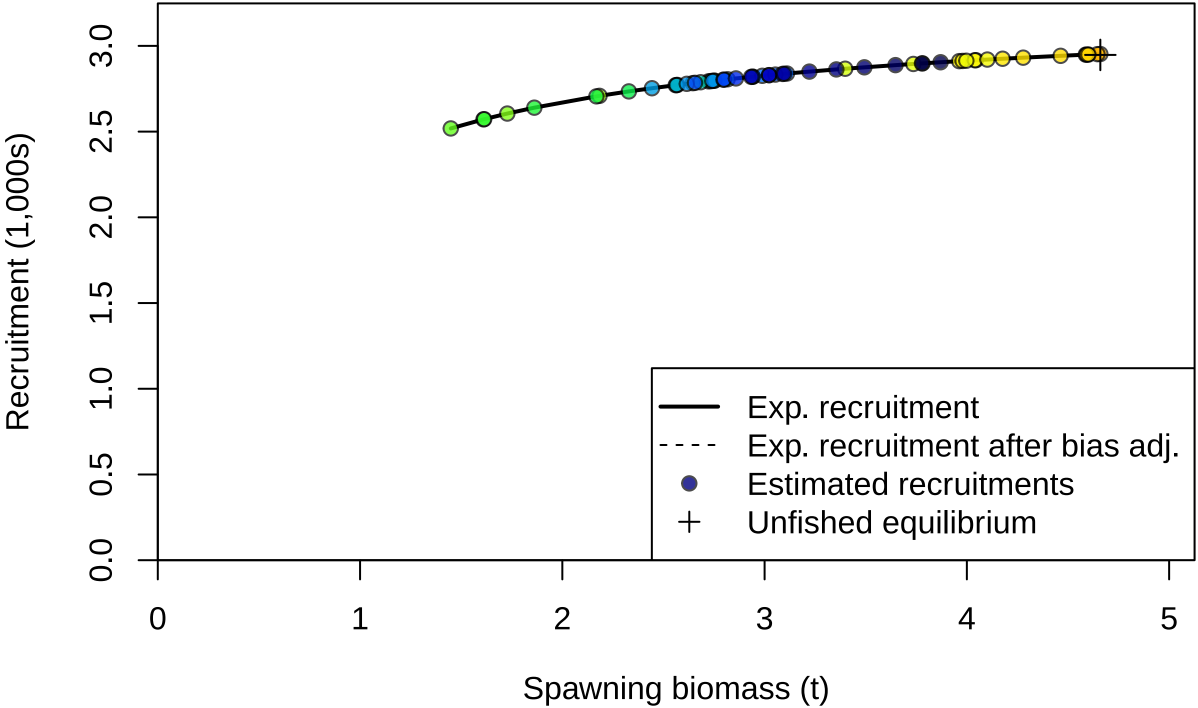
Age−0 recruits (1,000s)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
1970
1980
1990
2000
2010
2020
Year



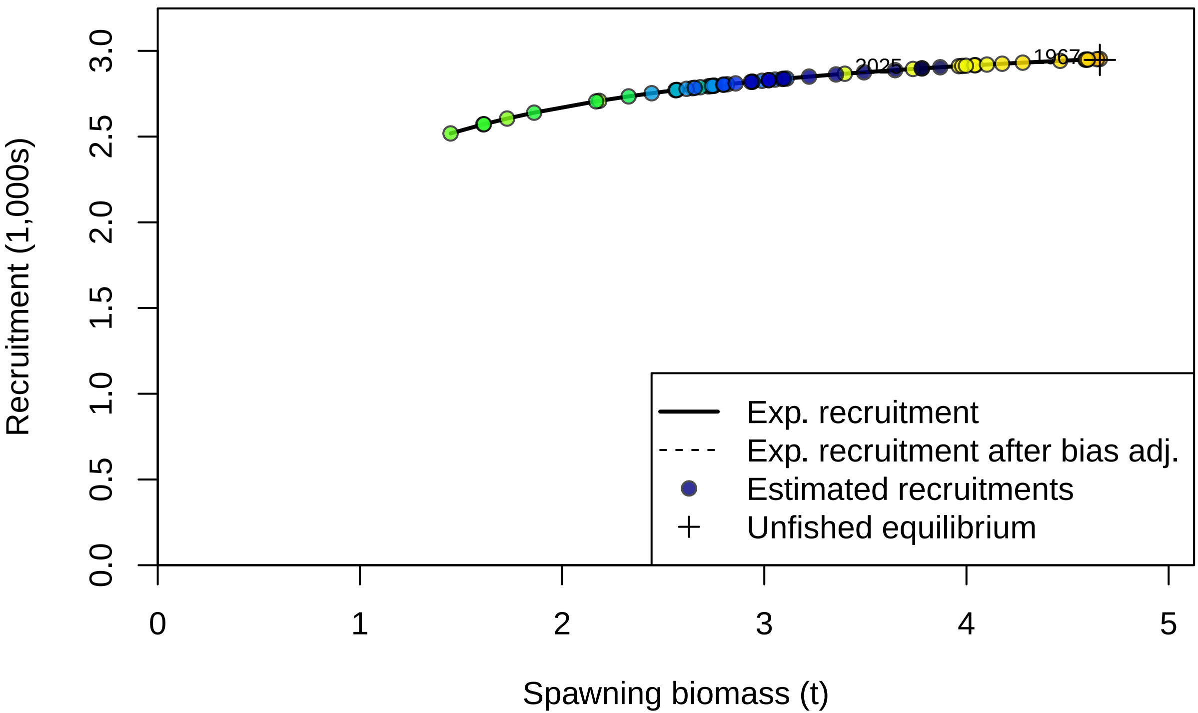
Age−0 recruits (1,000s)
0
1
2
3
4
5
1970
1980
1990
2000
2010
2020
Year


Summary Fishing Mortality
0.0
0.2
0.4
0.6
0.8
1970
1980
1990
2000
2010
2020
Year

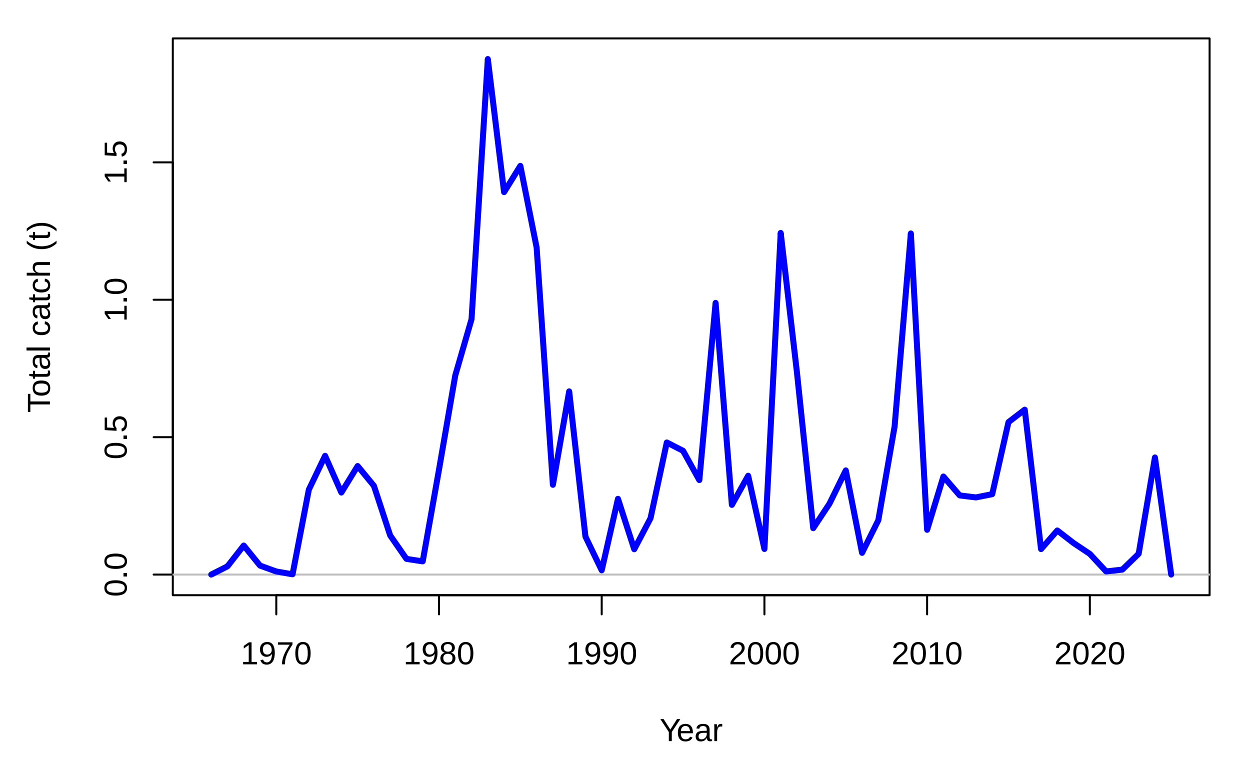

Spawning biomass (t)
0
1
2
3
4
5
equilibrium
without fishing
with fishing
1970
1980
1990
2000
2010
2020
Year



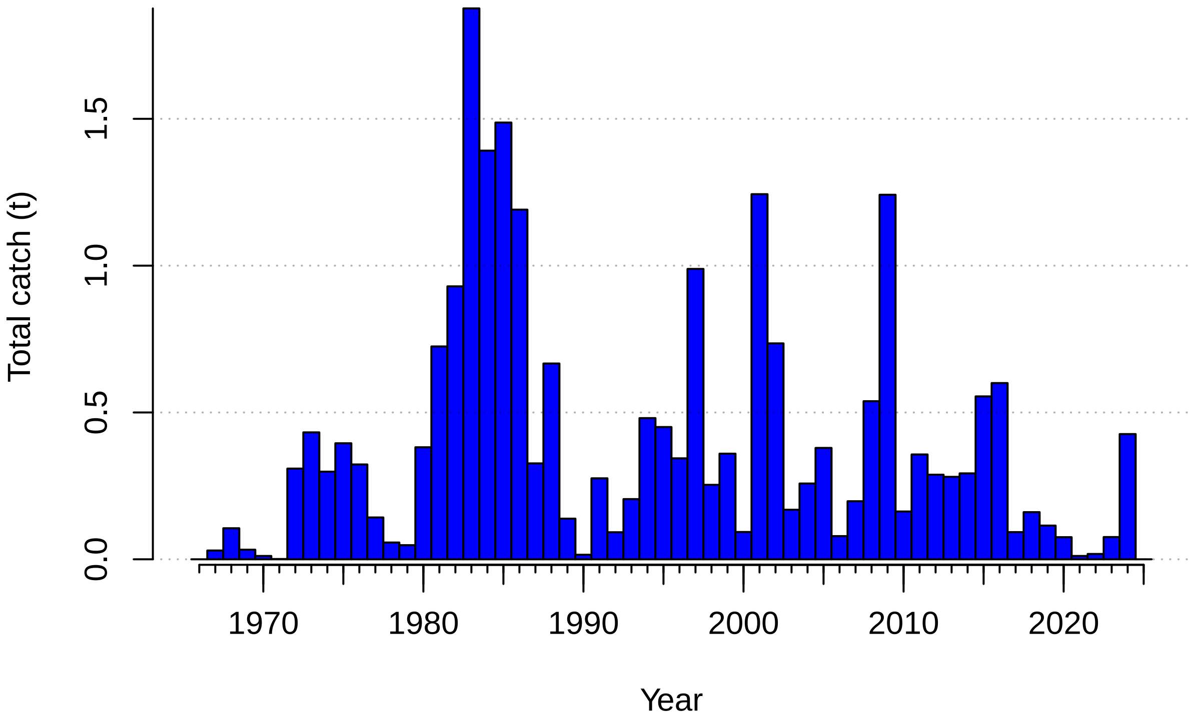
Recruitment (1,000s)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Spawning biomass (t)
0
1
2
3
4
5
Exp. recruitment
Exp. recruitment after bias adj.
Estimated recruitments
Unfished equilibrium



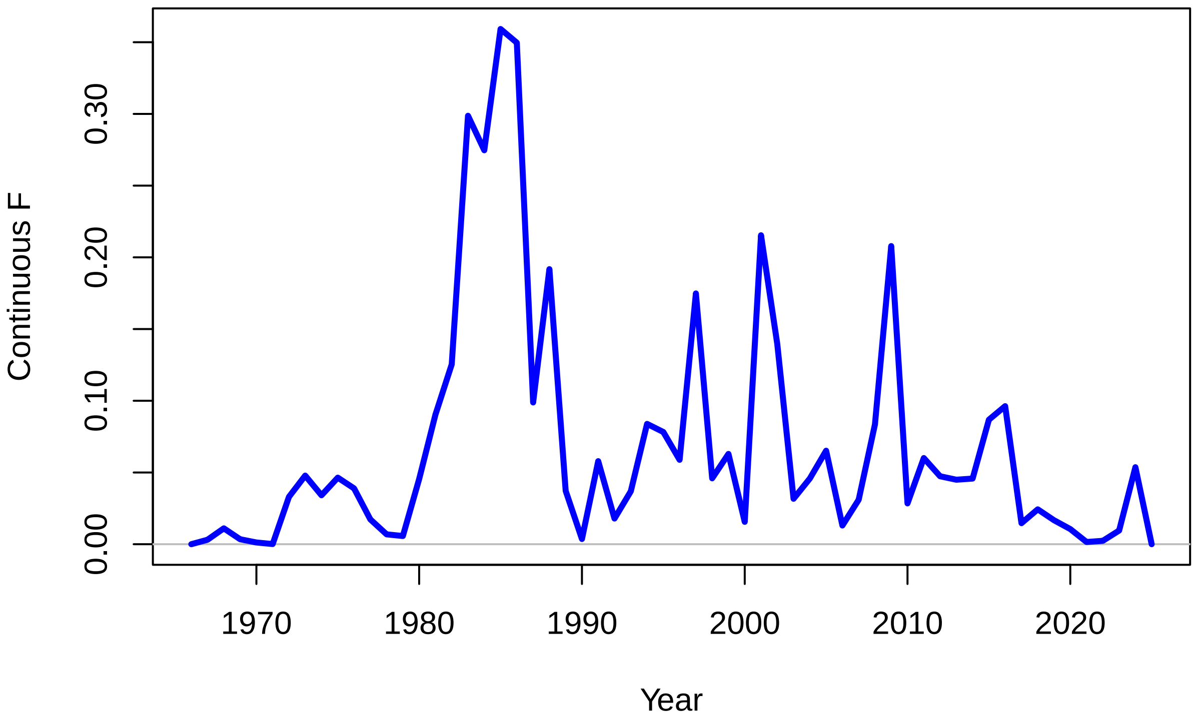
Recruitment (1,000s)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Spawning biomass (t)
0
1
2
3
4
5
2025
1967
Exp. recruitment
Exp. recruitment after bias adj.
Estimated recruitments
Unfished equilibrium



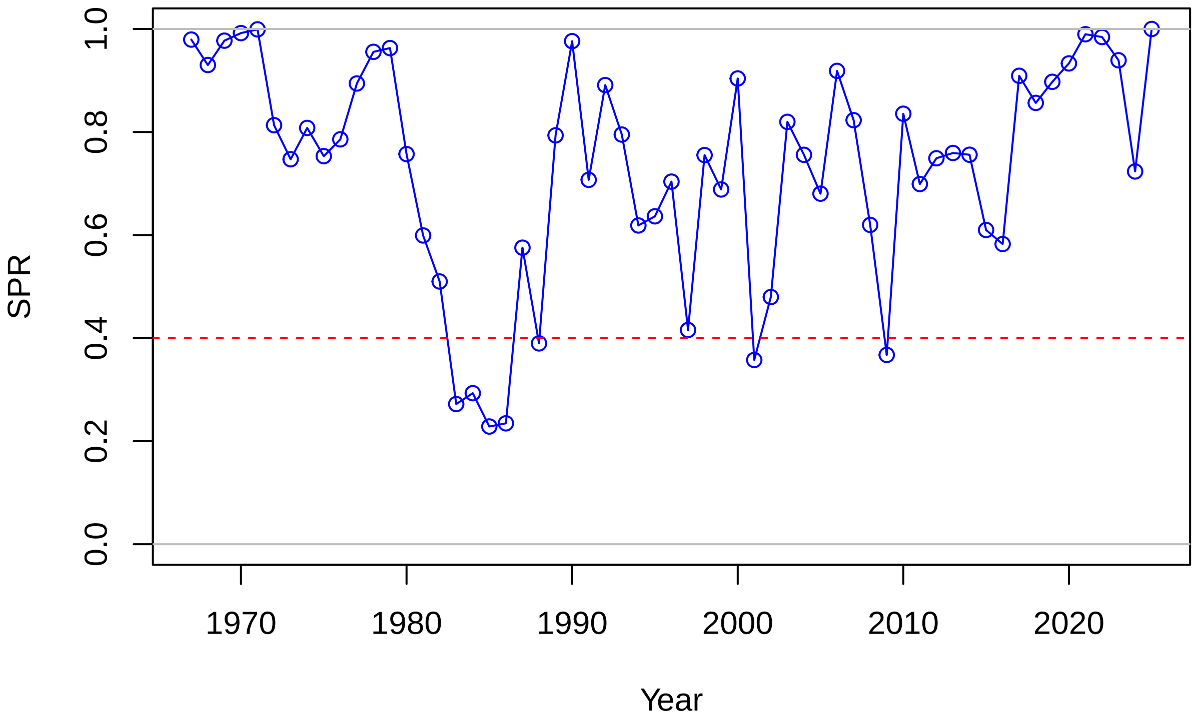
Total catch (t)
0.0
0.5
1.0
1.5
1970
1980
1990
2000
2010
2020
Year

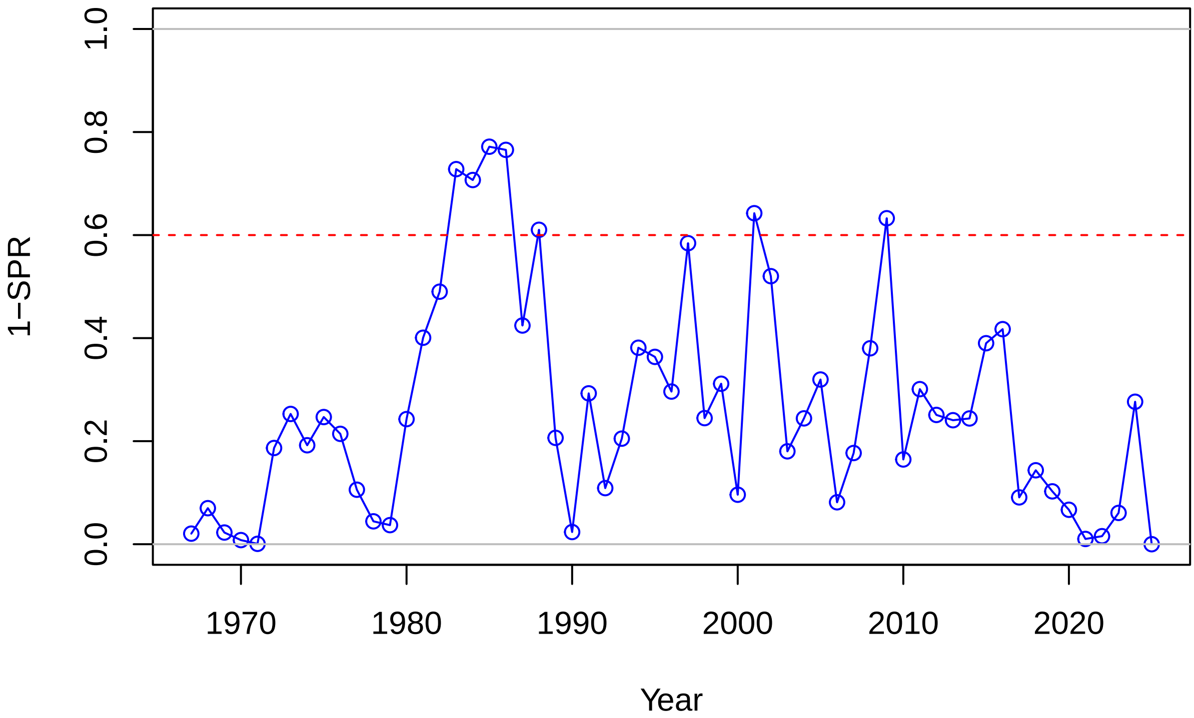


Total catch (t)
0.0
0.5
1.0
1.5
1970
1980
1990
2000
2010
2020
Year

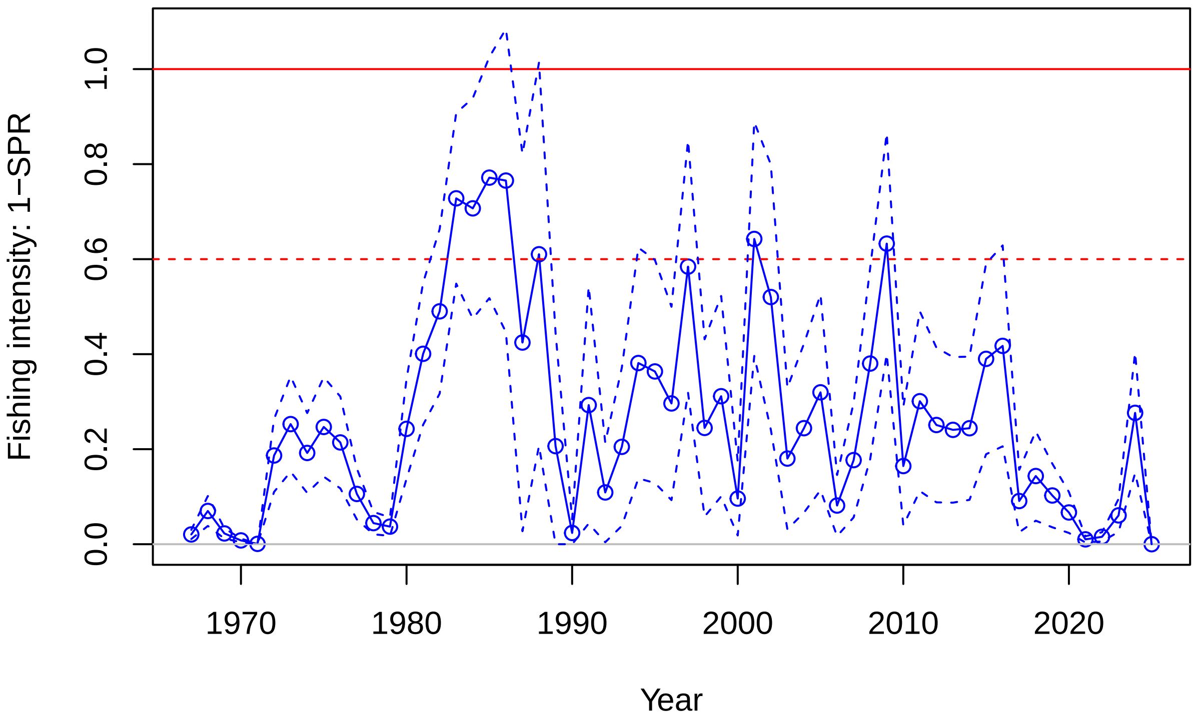


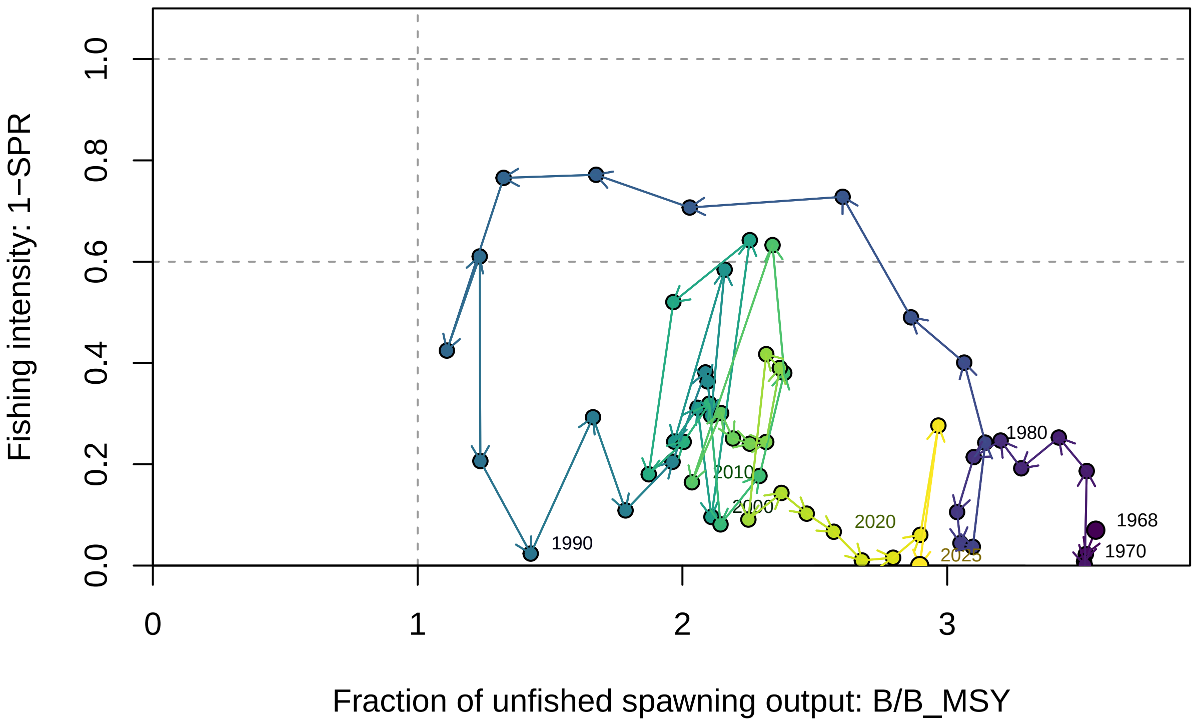
Continuous F
0.00
0.10
0.20
0.30
1970
1980
1990
2000
2010
2020
Year


SPR
1.0
0.8
0.6
0.4
0.2
0.0
1970
1980
1990
2000
2010
2020
Year

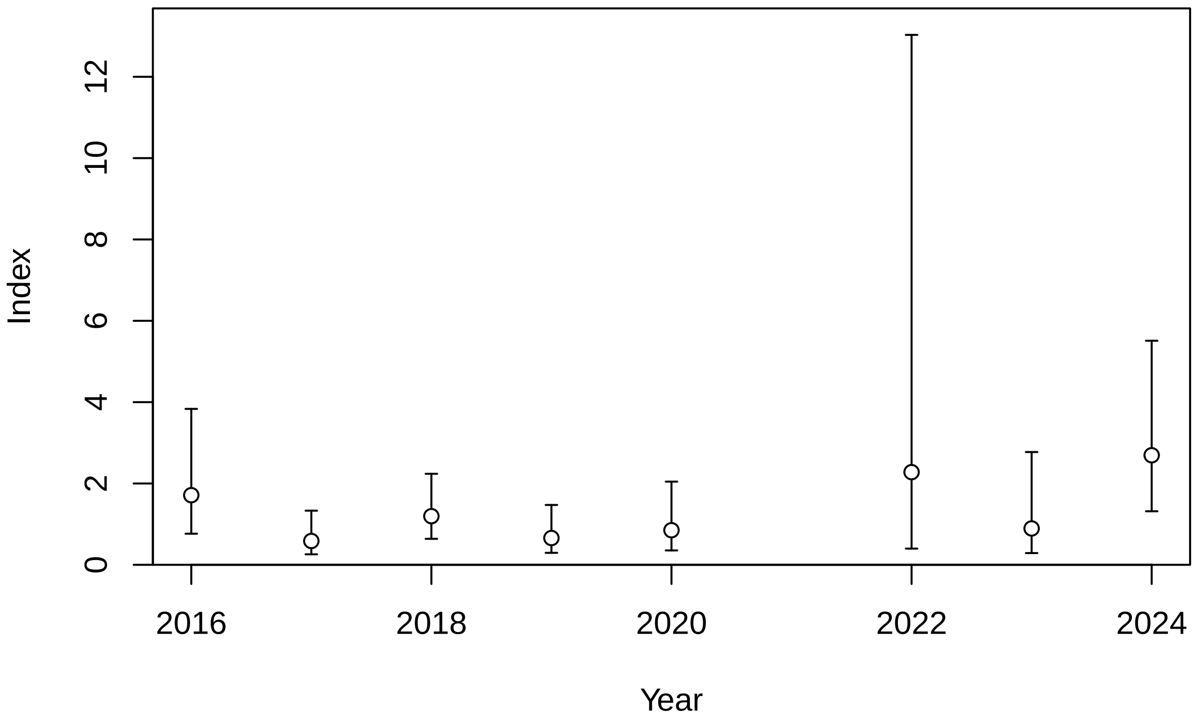


1−SPR
1.0
0.8
0.6
0.4
0.2
0.0
1970
1980
1990
2000
2010
2020
Year

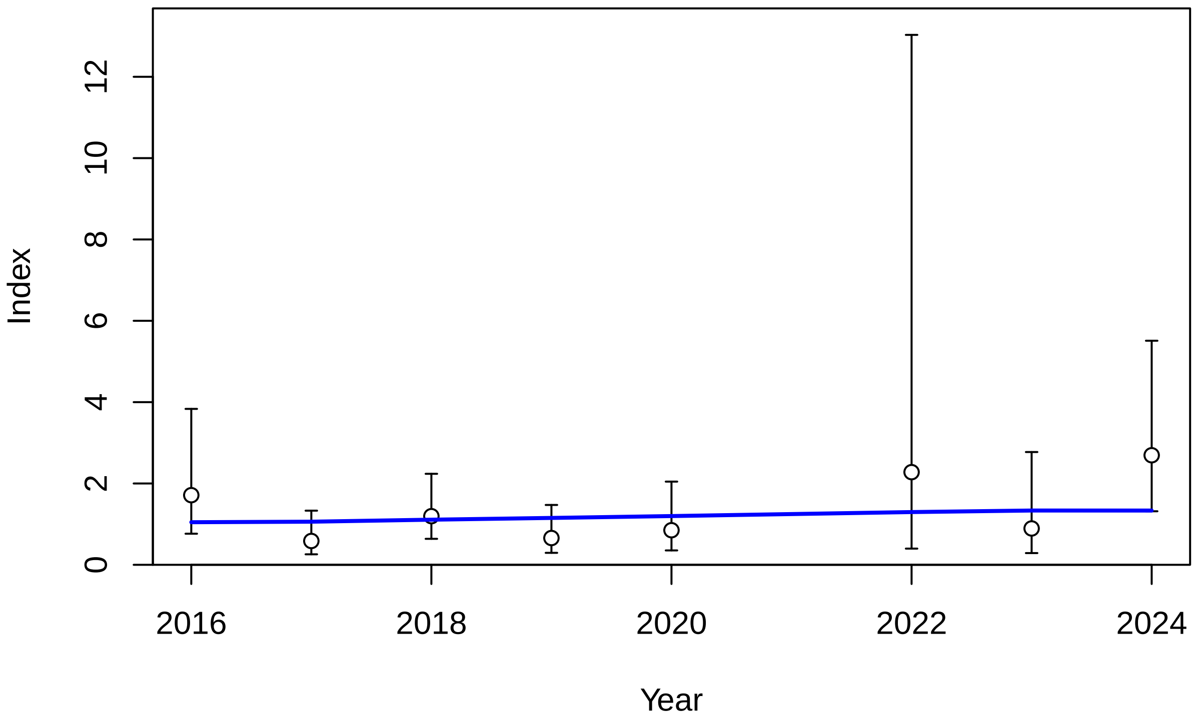

Fishing intensity: 1−SPR
0.0
0.2
0.4
0.6
0.8
1.0
1970
1980
1990
2000
2010
2020
Year



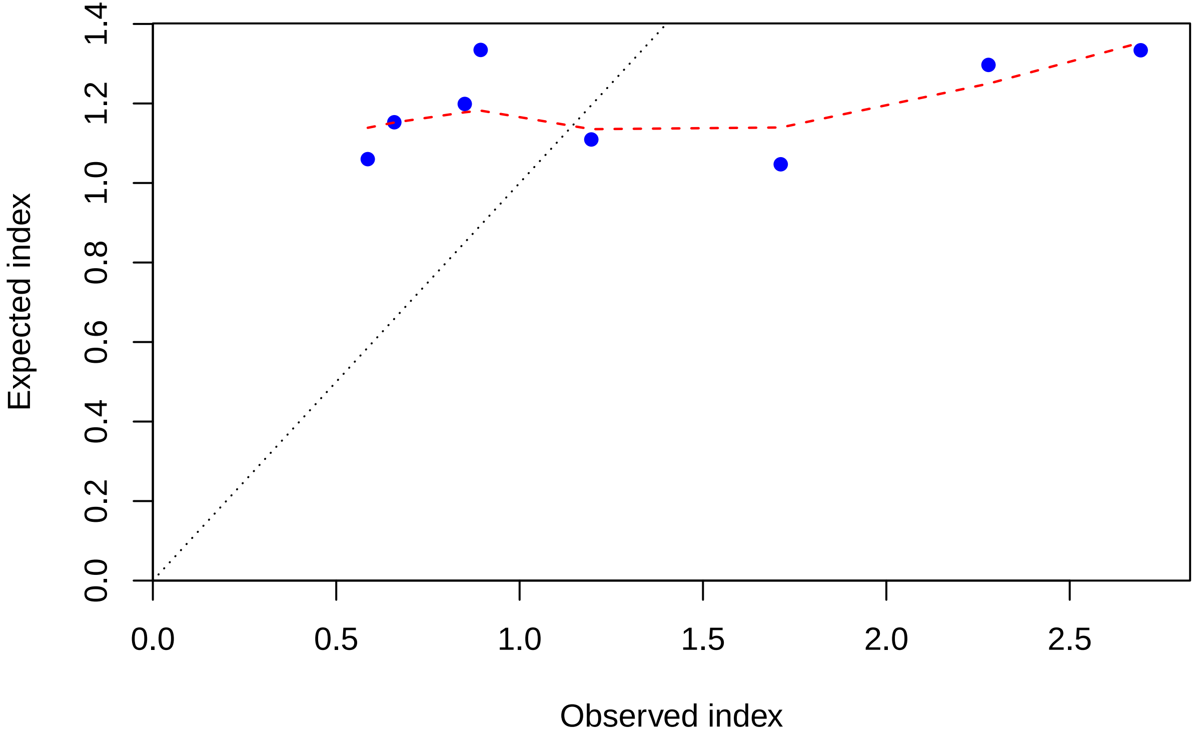
Fishing intensity: 1−SPR
Fraction of unfished spawning output: B/B_MSY
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
1968
1970
1980
1990
2000
2010
2020
2025



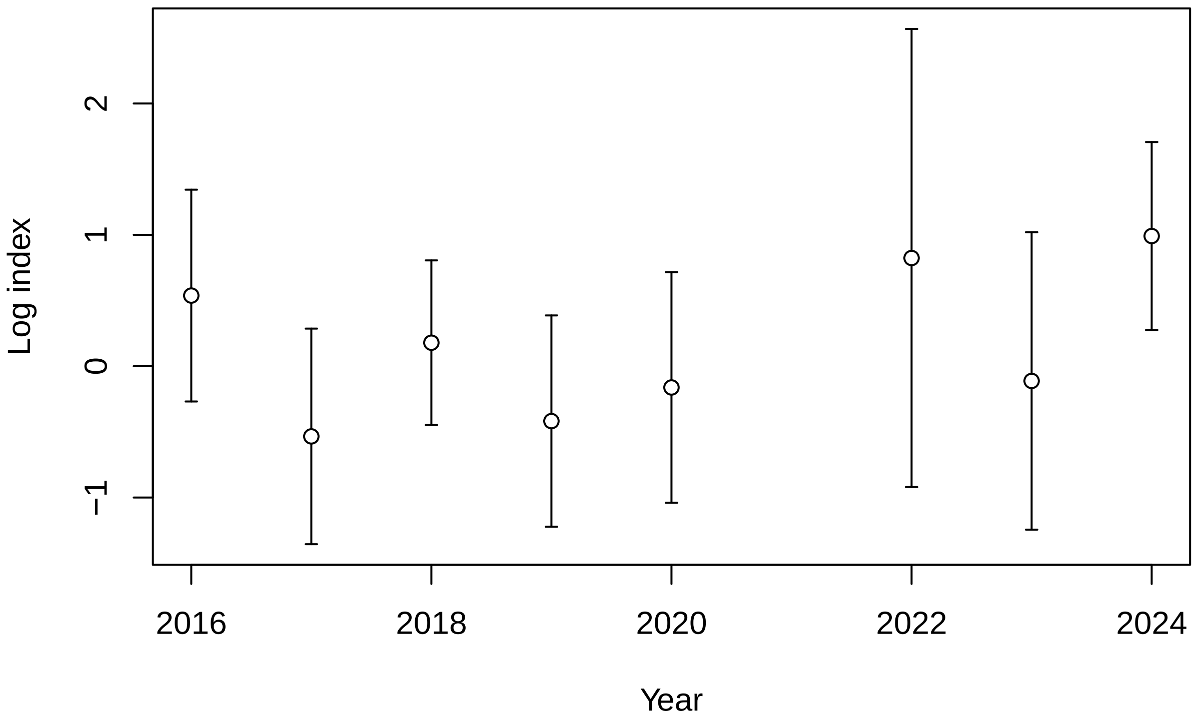
Index
0
2
4
6
8
10
12
2016
2018
2020
2022
2024
Year

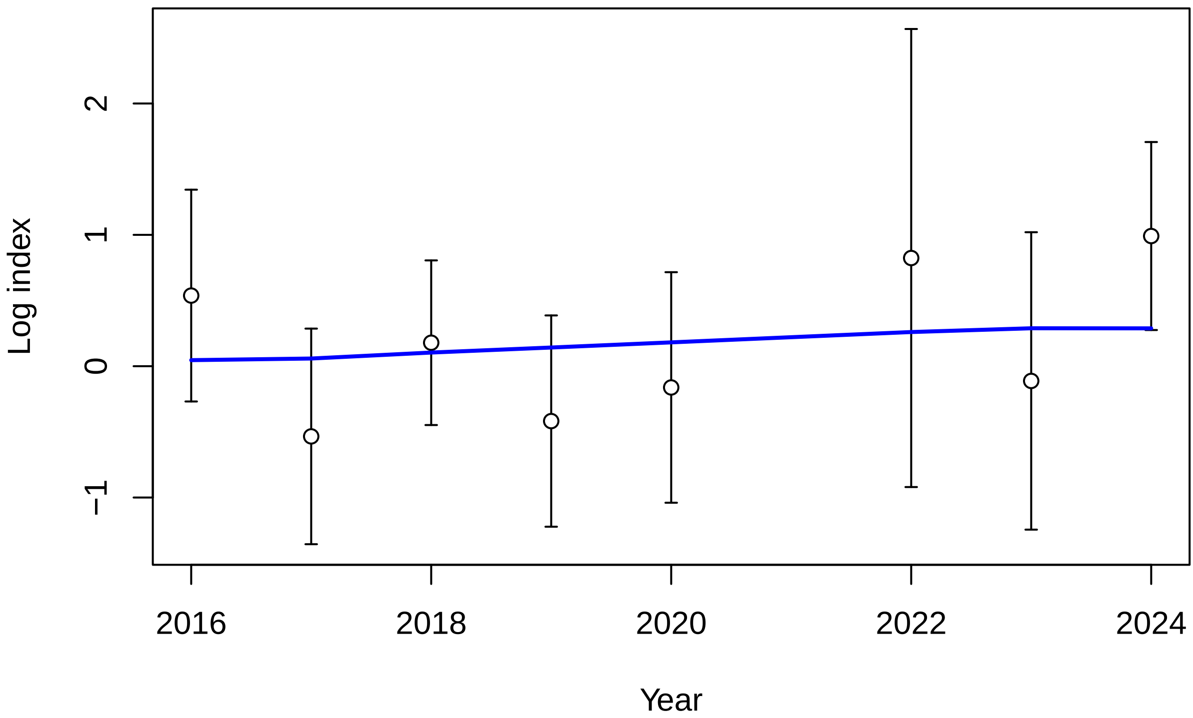


Index
0
2
4
6
8
10
12
2016
2018
2020
2022
2024
Year

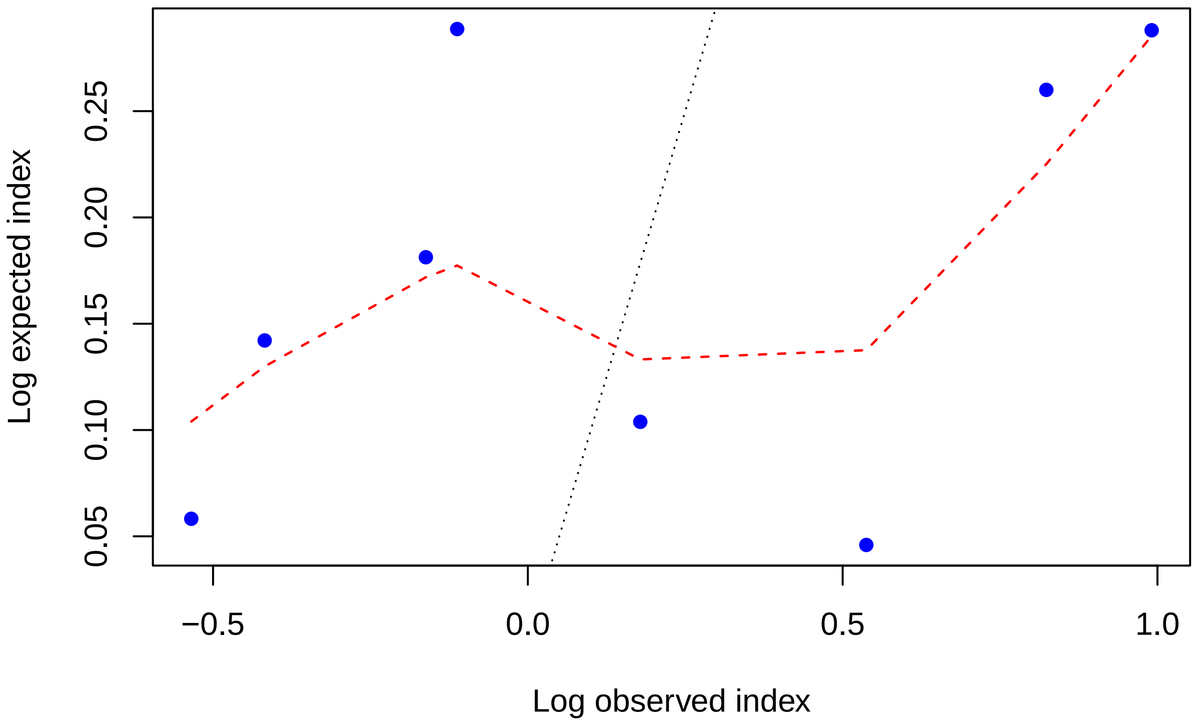


Expected index
Observed index
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0.0
0.5
1.0
1.5
2.0
2.5

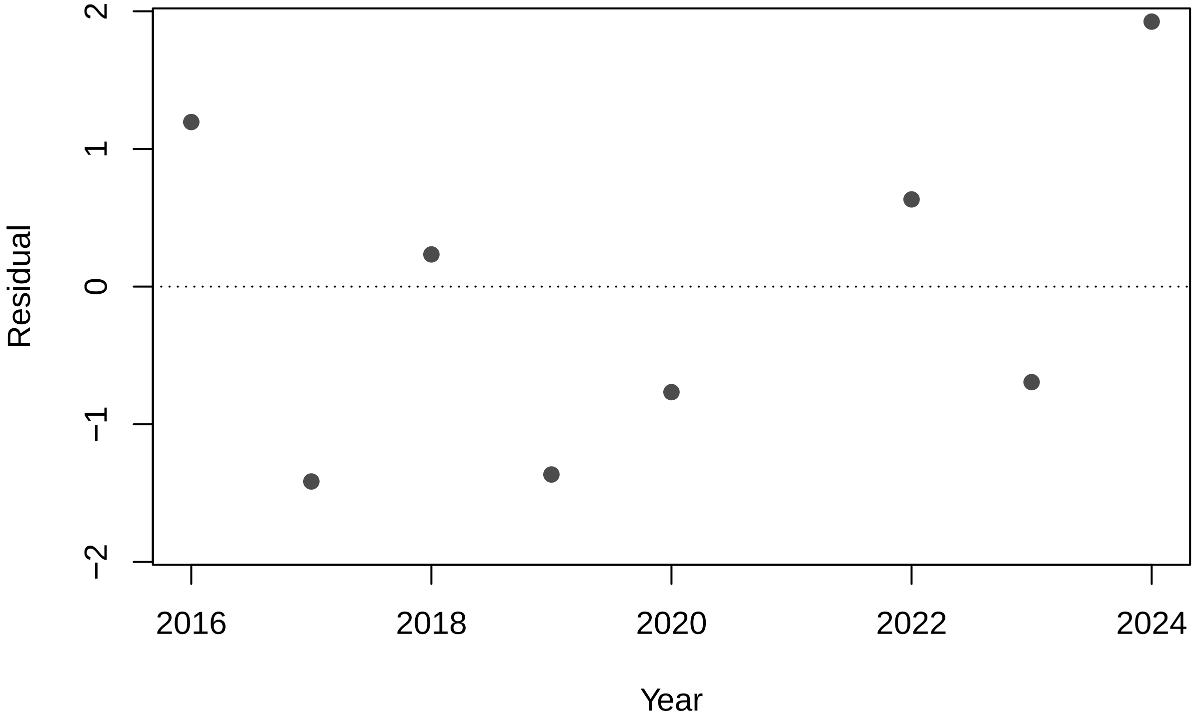

Log index
2
1
0
−1
2016
2018
2020
2022
2024
Year

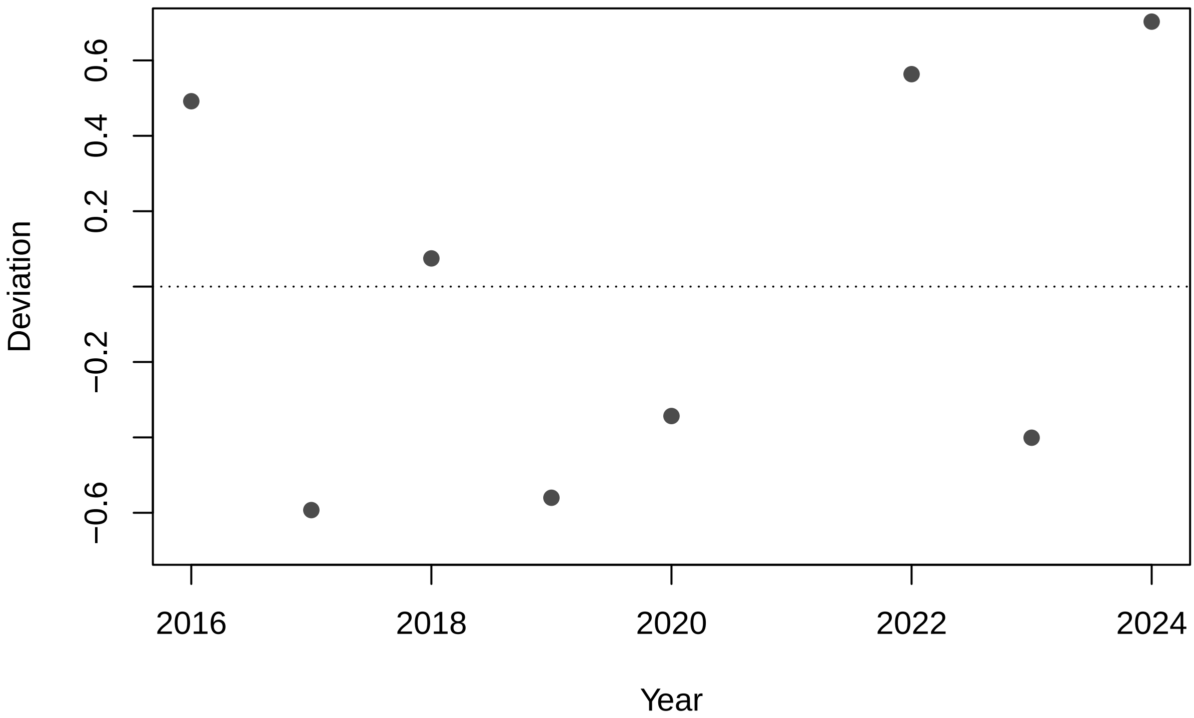


Log index
−1
0
1
2
2016
2018
2020
2022
2024
Year

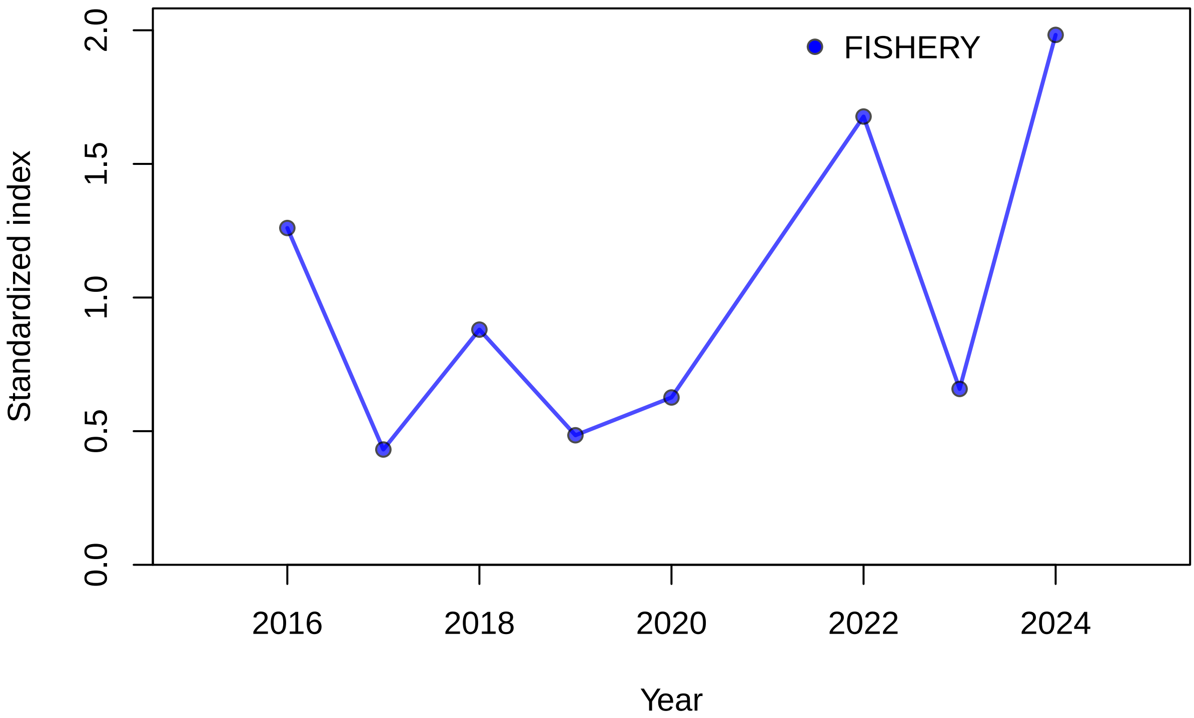


Log expected index
0.05
0.10
0.15
0.20
0.25
−0.5
0.0
0.5
1.0
Log observed index

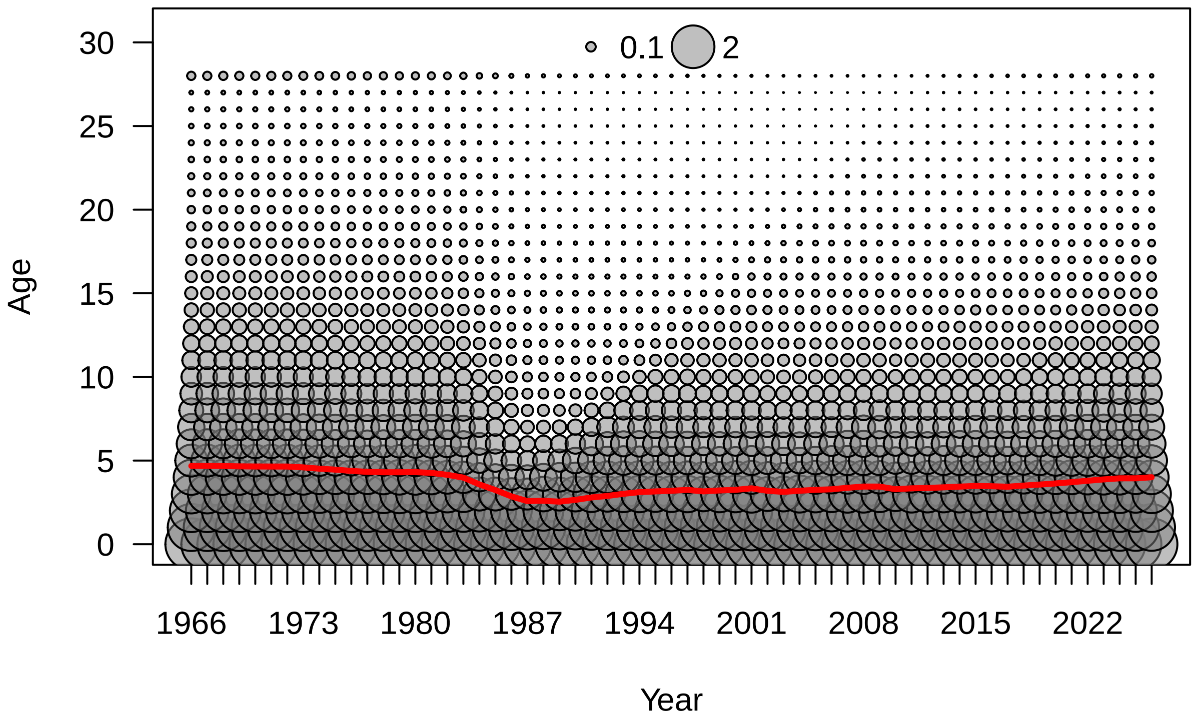

Residual
2
1
0
−1
−2
2016
2018
2020
2022
2024
Year



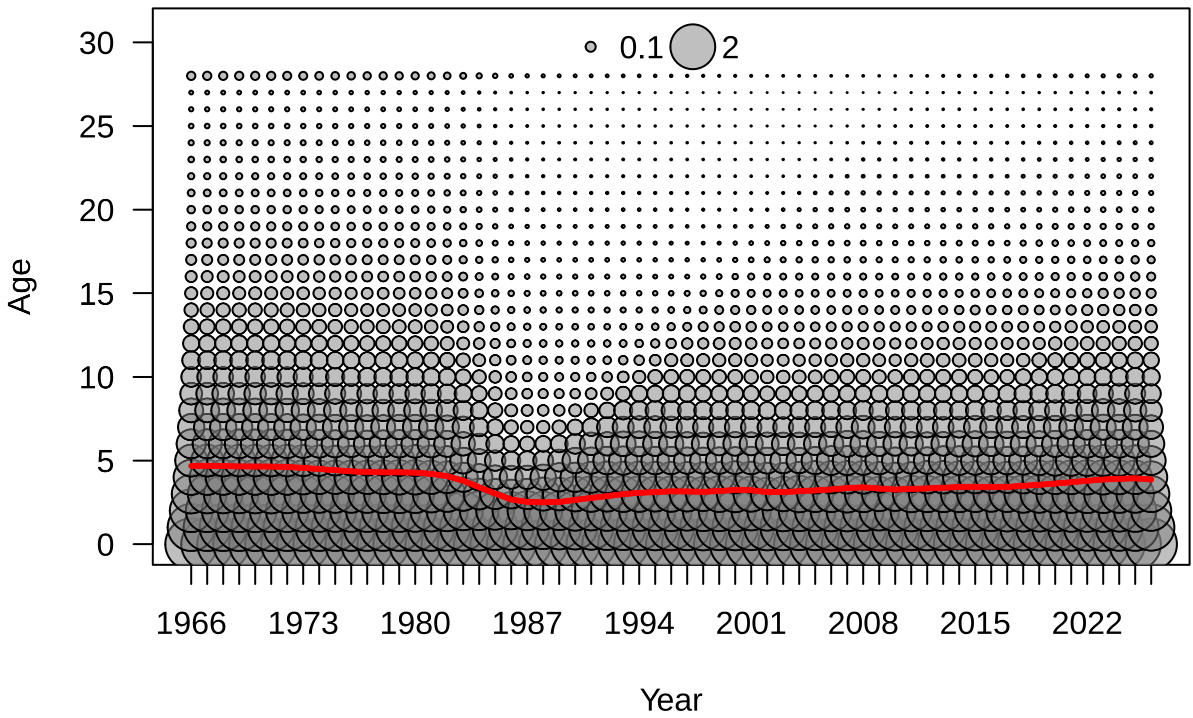
Deviation
0.6
0.4
0.2
−0.2
−0.6
2016
2018
2020
2022
2024
Year



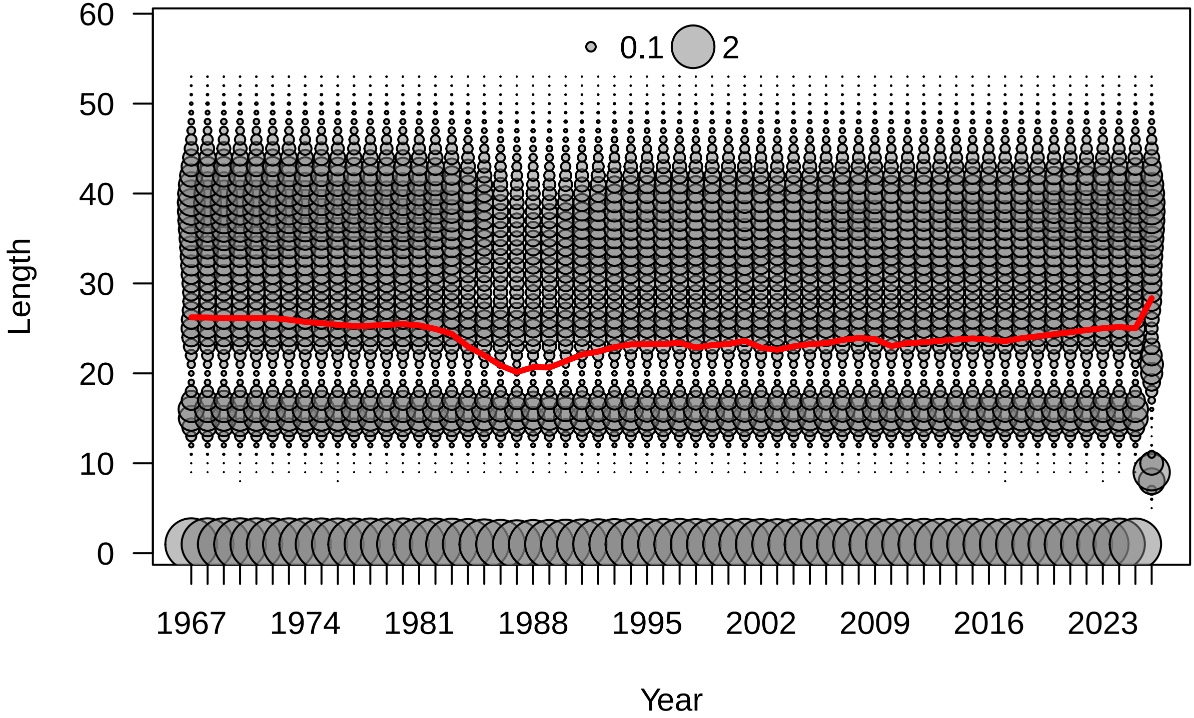
FISHERY
Standardized index
0.0
0.5
1.0
1.5
2.0
2016
2018
2020
2022
2024
Year



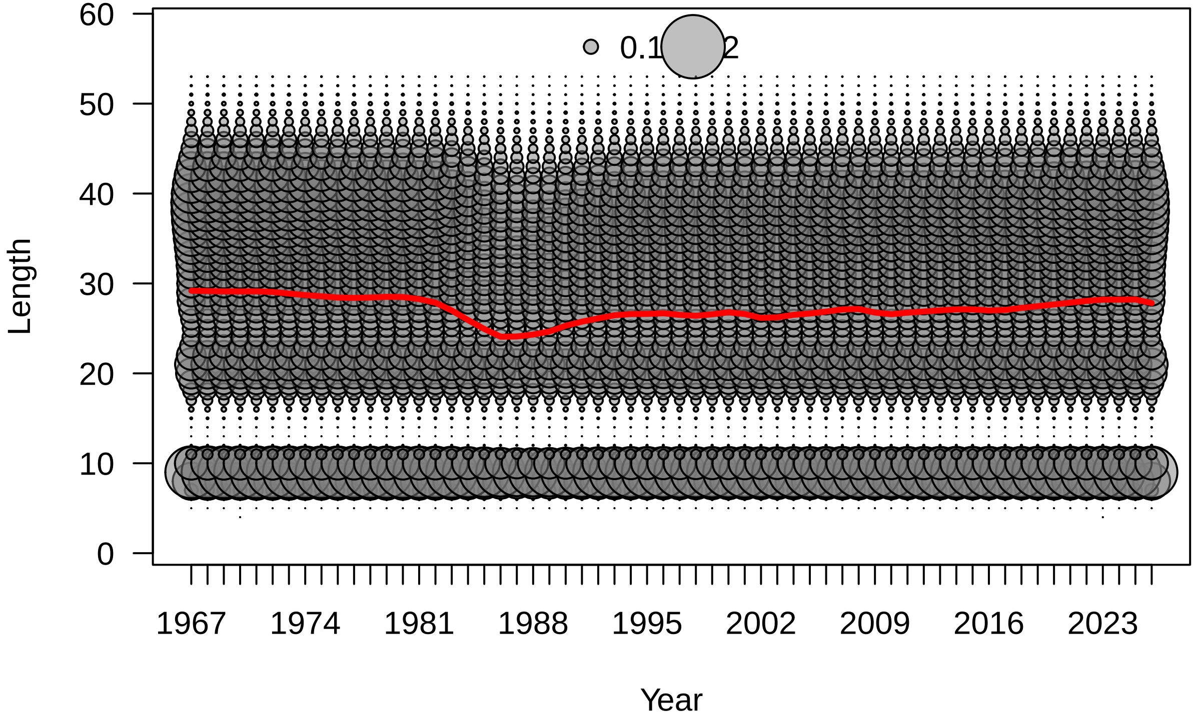
0.1
2
30
25
20
15
10
5
0
Age
1966
1973
1980
1987
1994
2001
2008
2015
2022
Year



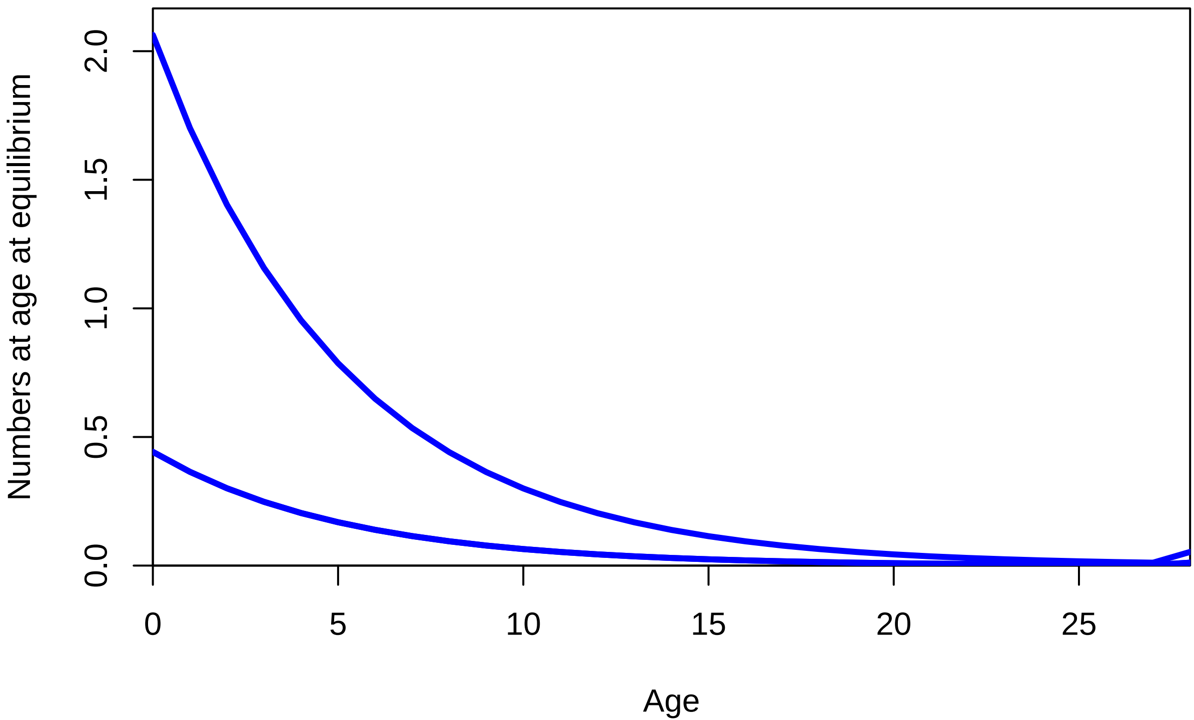
0.1
2
30
25
20
15
10
5
0
Age
1966
1973
1980
1987
1994
2001
2008
2015
2022
Year



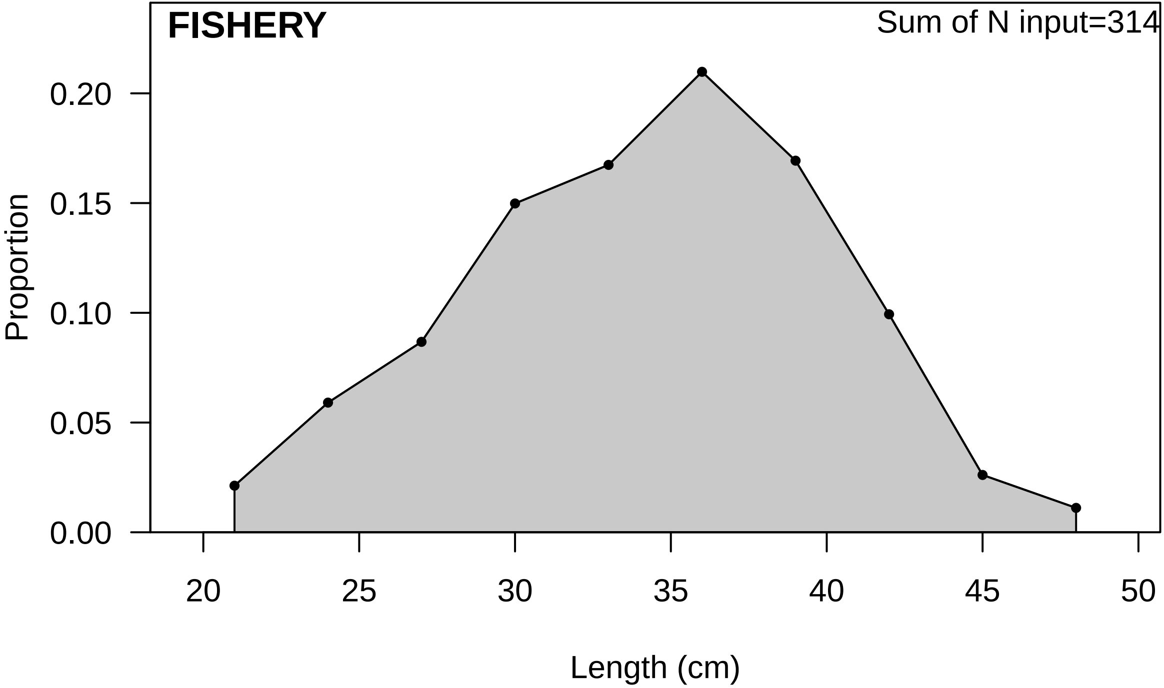
0.1
2
60
50
40
30
20
10
0
Length
1967
1974
1981
1988
1995
2002
2009
2016
2023
Year

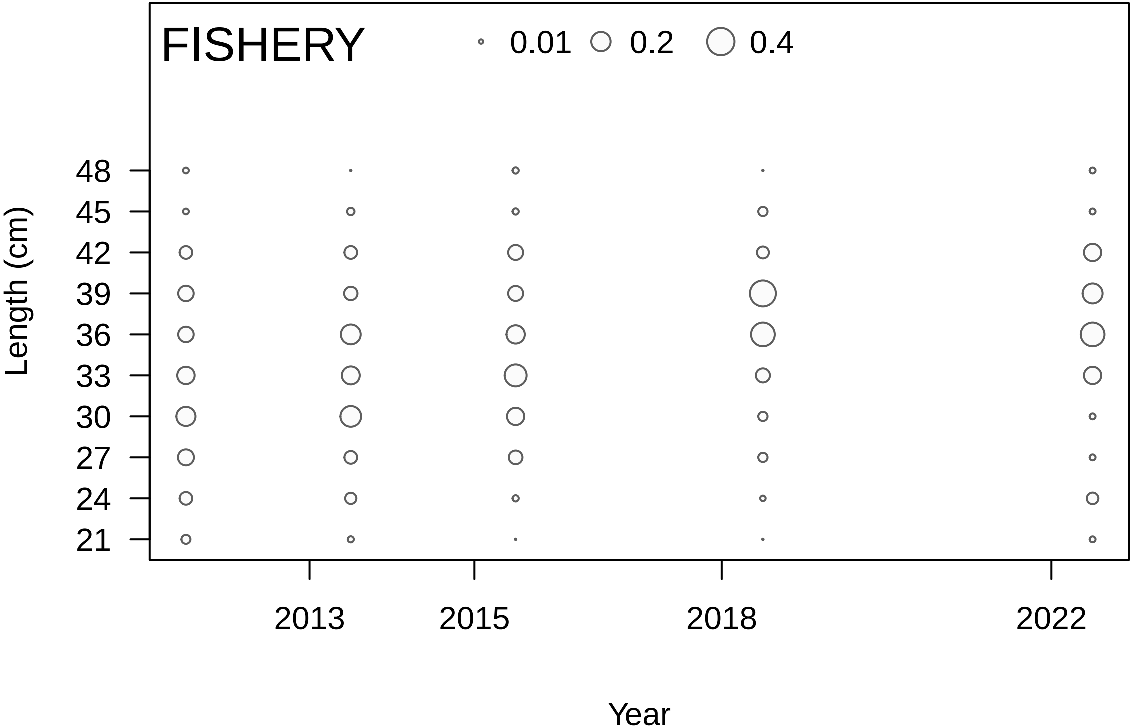


60
50
40
30
20
10
0
Length
0.1
2
1967
1974
1981
1988
1995
2002
2009
2016
2023
Year

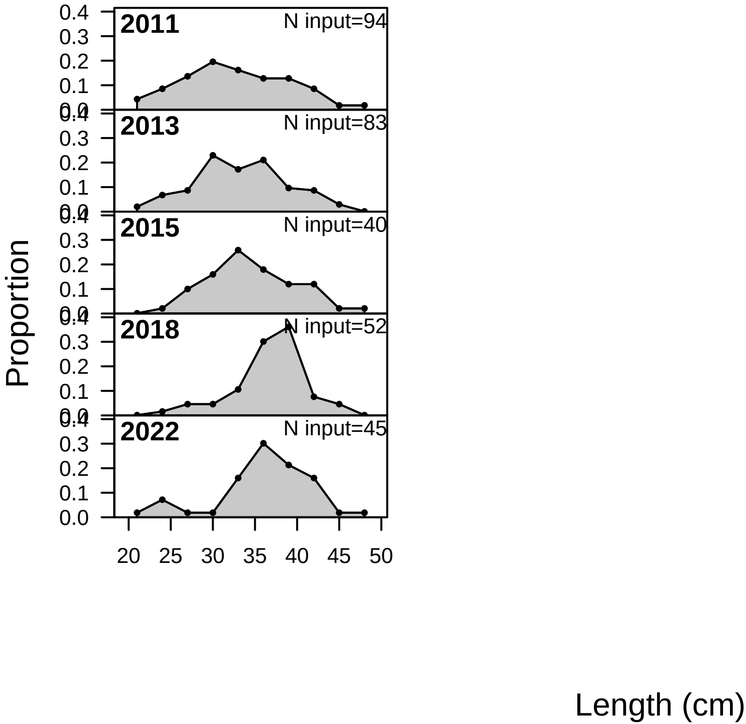


Numbers at age at equilibrium
0.0
0.5
1.0
1.5
2.0
0
5
10
15
20
25
Age

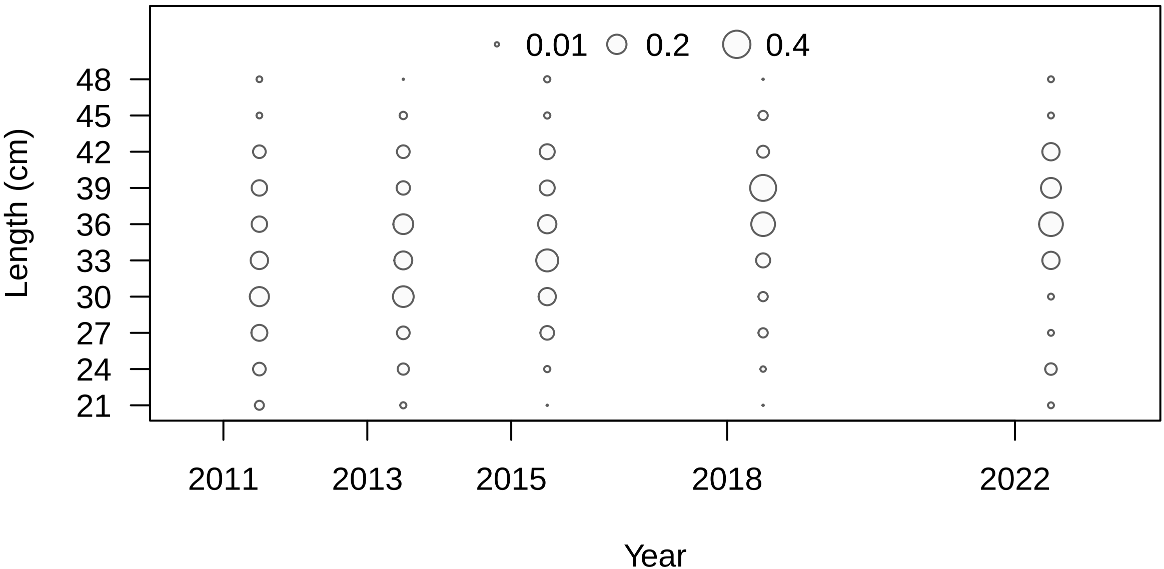

FISHERY
Sum of N input=314
Proportion
0.00
0.05
0.10
0.15
0.20
20
25
30
35
40
45
50
Length (cm)



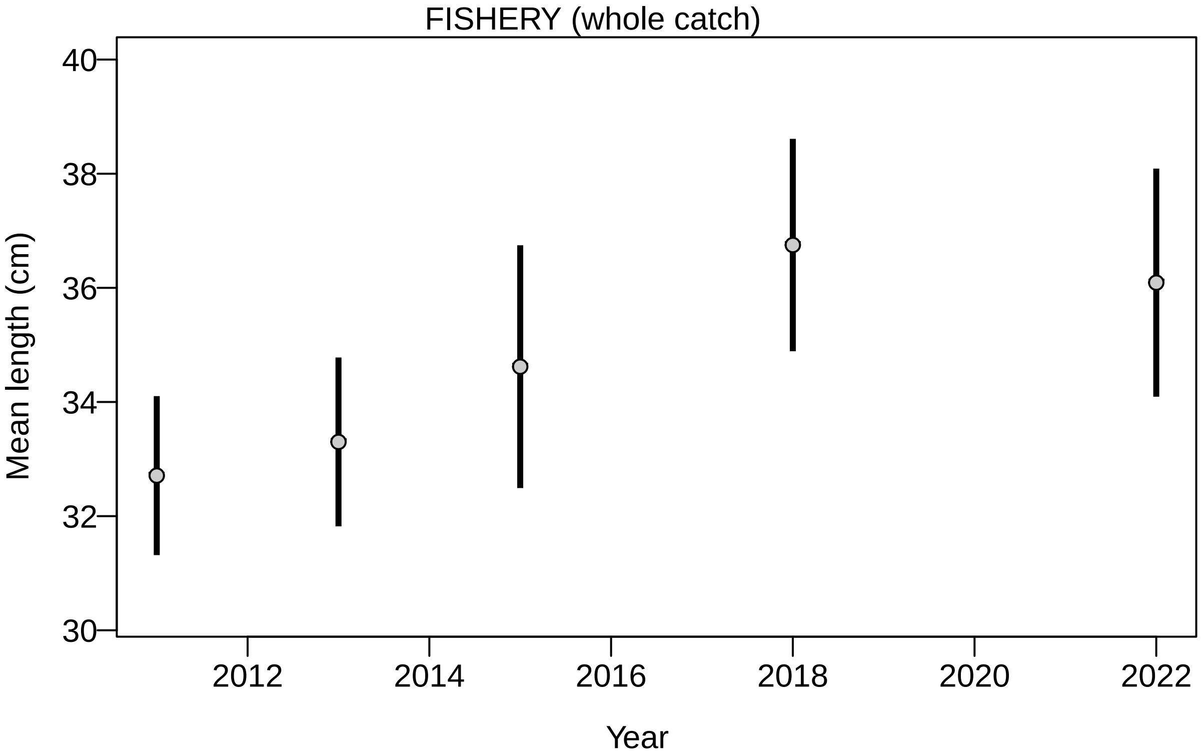
FISHERY
0.01
0.2
0.4
48
45
42
39
36
33
30
27
24
21
Length (cm)
2013
2015
2018
2022
Year



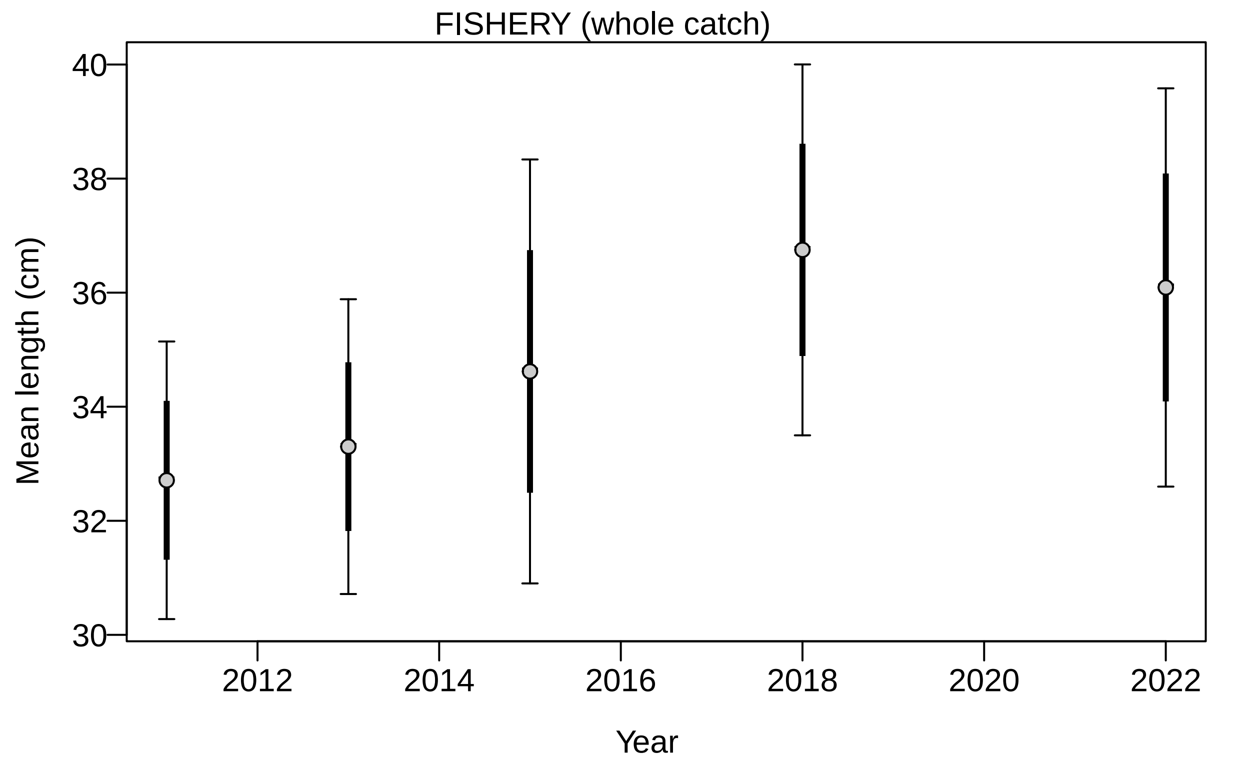
2011
N input=94
2013
N input=83
2015
N input=40
2018
N input=52
2022
N input=45
Proportion
0.0
0.1
0.2
0.3
0.4
20
25
30
35
40
45
50
Length (cm)

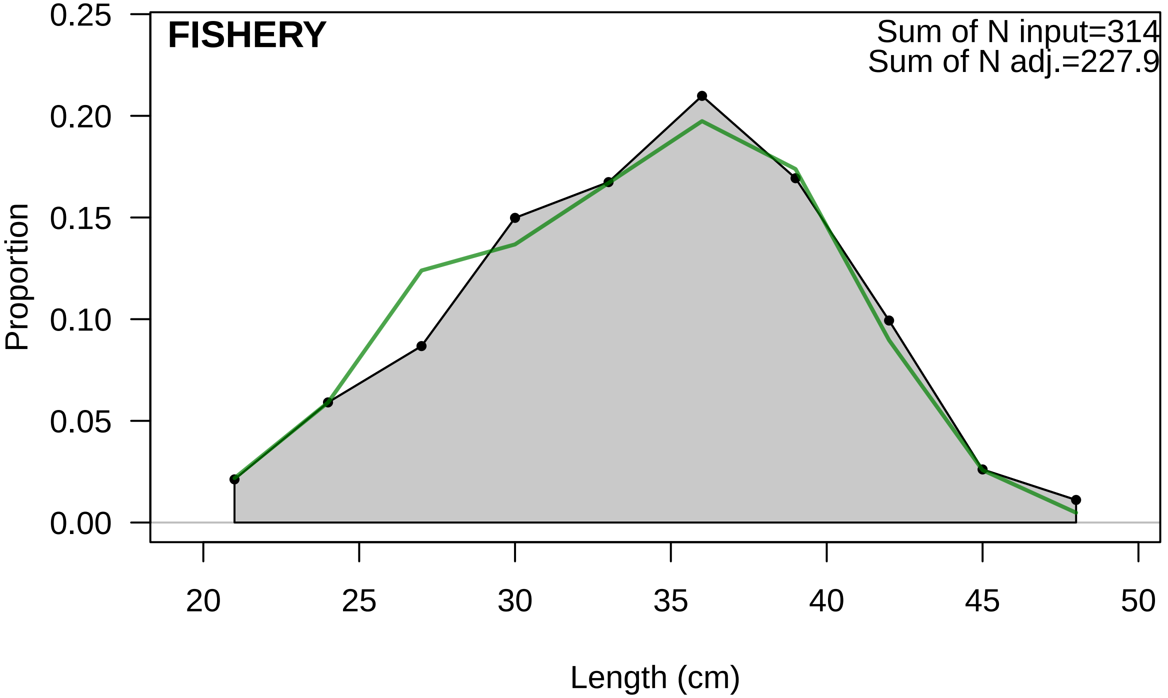


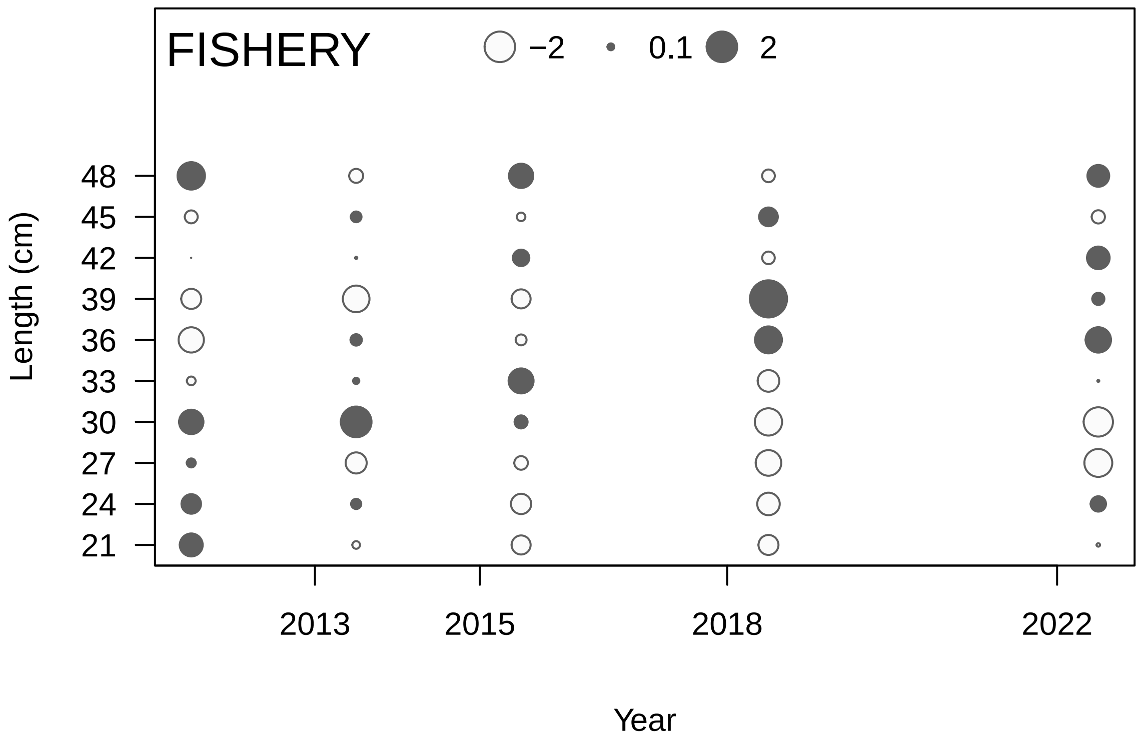
0.01
0.2
0.4
Length (cm)
48
45
42
39
36
33
30
27
24
21
2011
2013
2015
2018
2022
Year

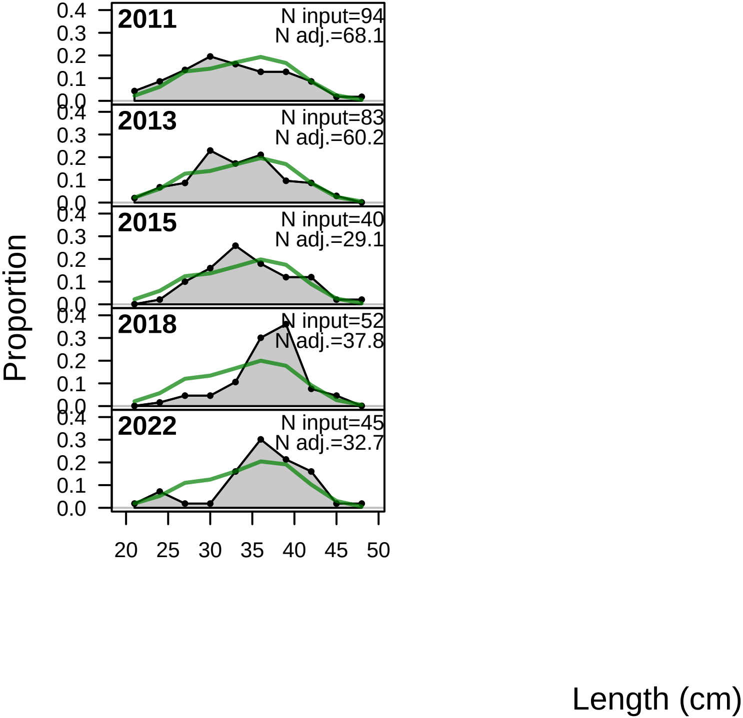


FISHERY (whole catch)
Mean length (cm)
40
38
36
34
32
30
2012
2014
2016
2018
2020
2022
Year

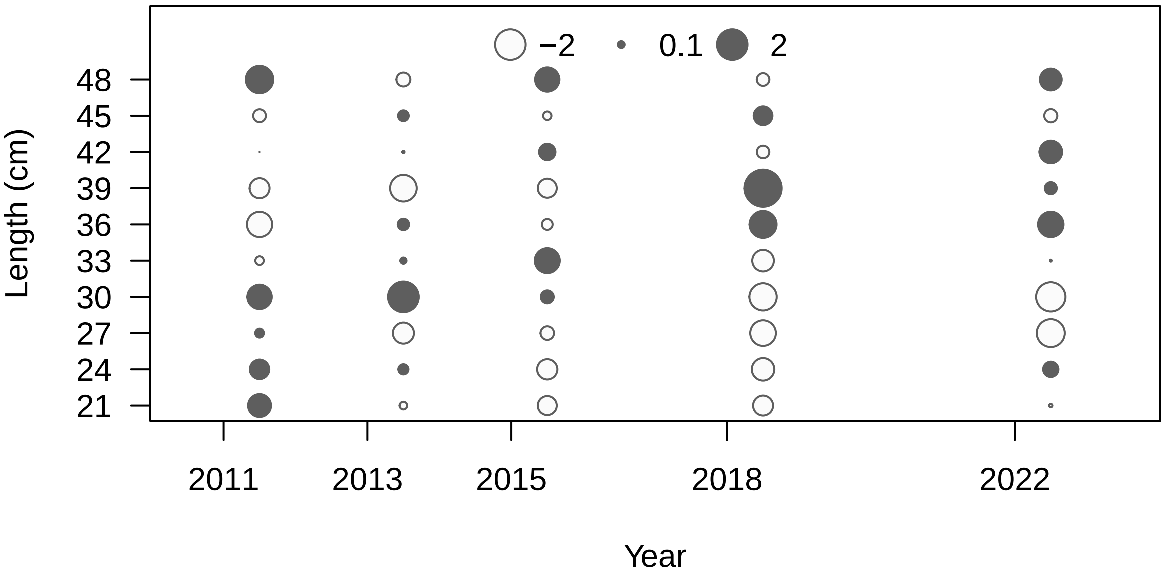

FISHERY (whole catch)
Mean length (cm)
30
32
34
36
38
40
2012
2014
2016
2018
2020
2022
Year

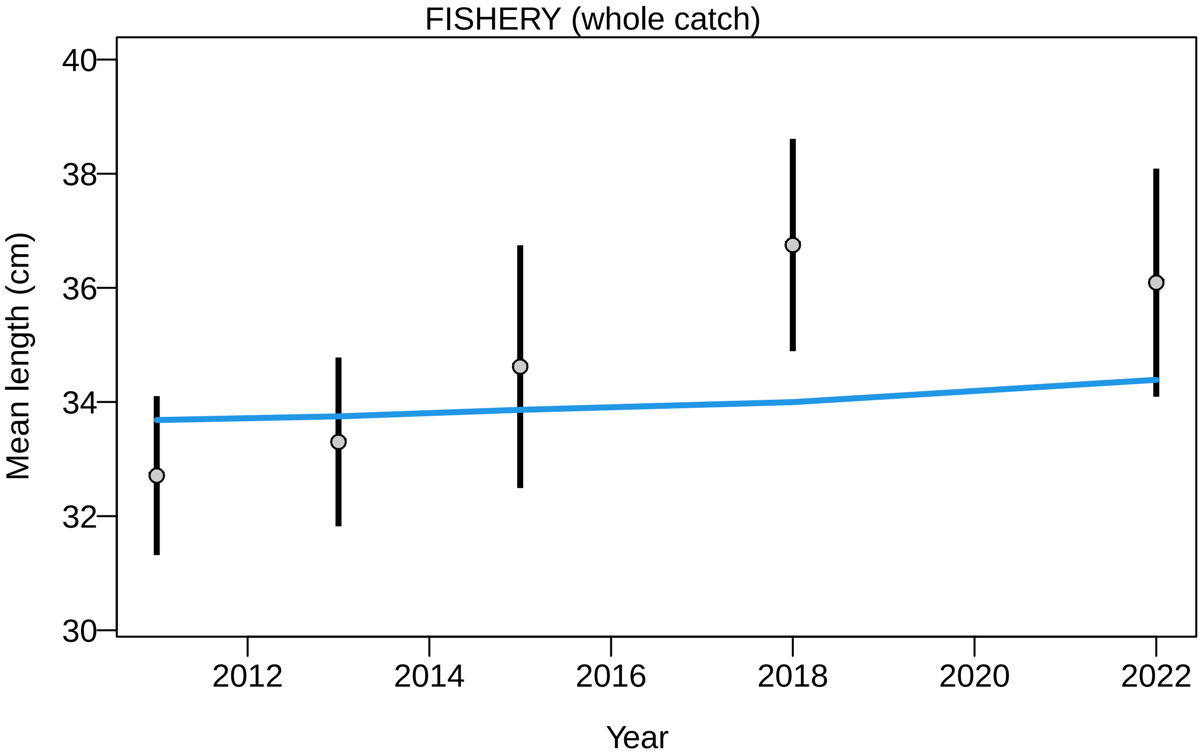

FISHERY
Sum of N input=314
Sum of N adj.=227.9
Proportion
0.25
0.20
0.15
0.10
0.05
0.00
20
25
30
35
40
45
50
Length (cm)

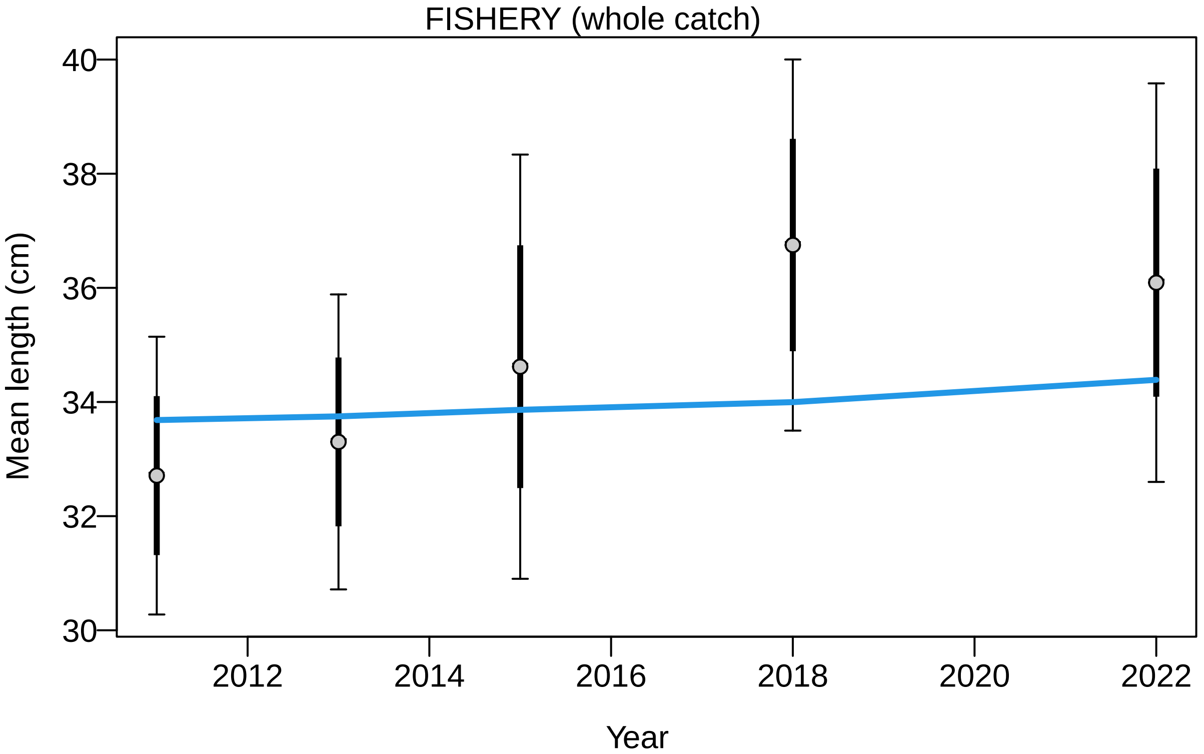

FISHERY
−2
0.1
2
48
45
42
39
36
33
30
27
24
21
Length (cm)
2013
2015
2018
2022
Year



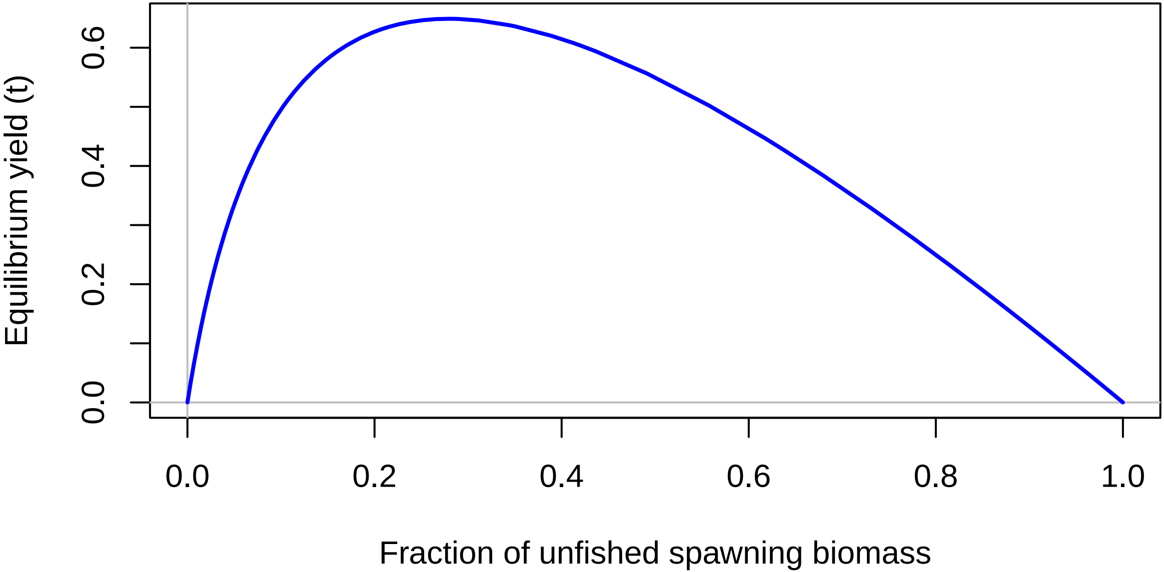
2011
N input=94
N adj.=68.1
2013
N input=83
N adj.=60.2
2015
N input=40
N adj.=29.1
2018
N input=52
N adj.=37.8
2022
N input=45
N adj.=32.7
0.0
0.1
0.2
0.3
0.4
20
25
30
35
40
45
50
Proportion
Length (cm)


−2
0.1
2
Length (cm)
48
45
42
39
36
33
30
27
24
21
2011
2013
2015
2018
2022
Year

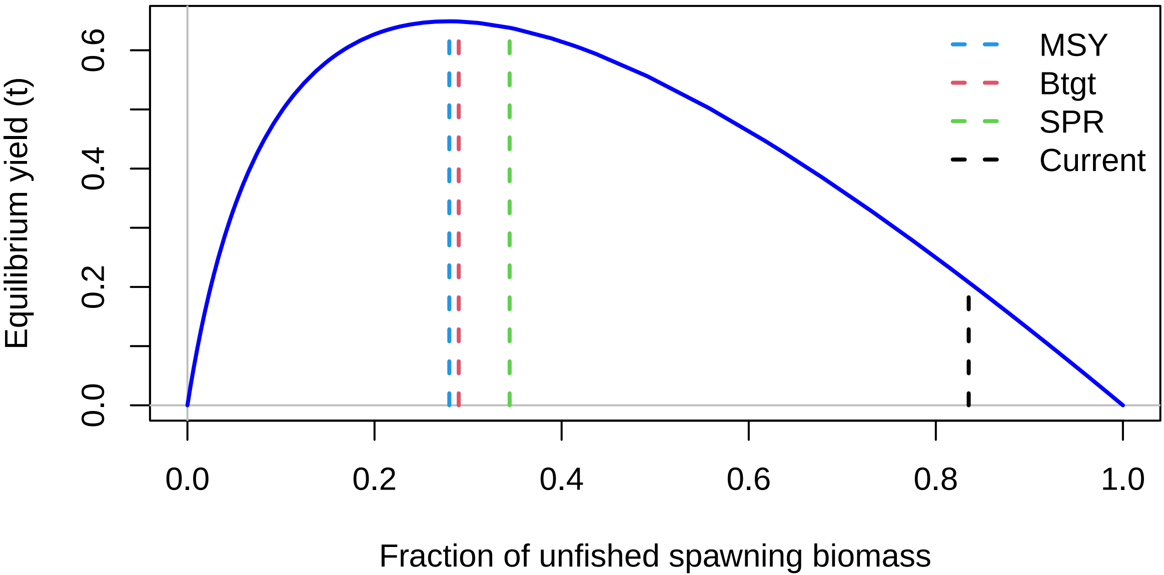

FISHERY (whole catch)
Mean length (cm)
40
38
36
34
32
30
2012
2014
2016
2018
2020
2022
Year

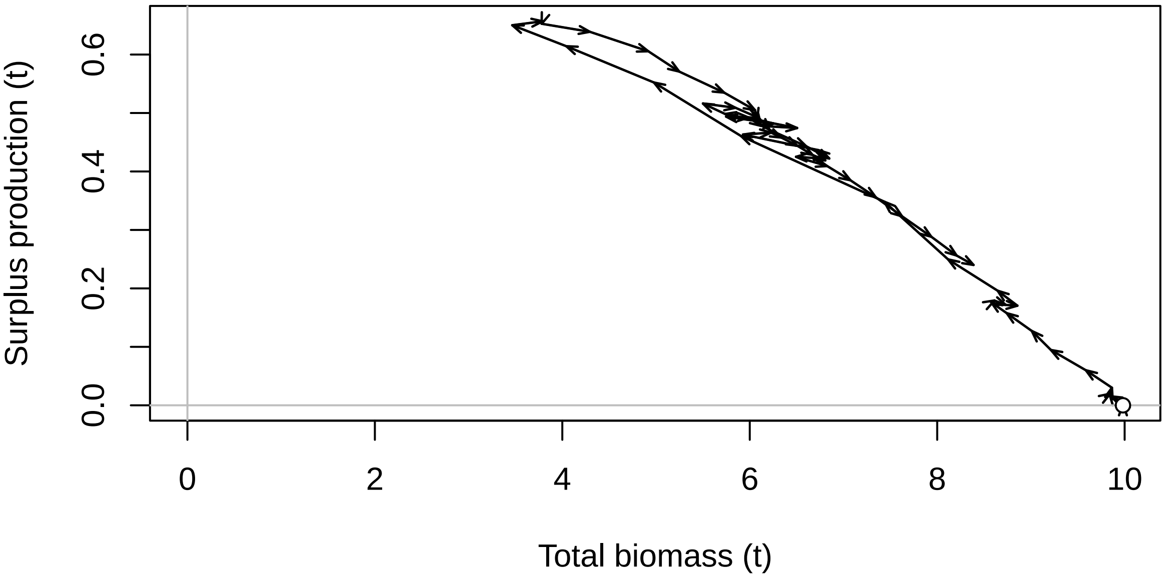


FISHERY (whole catch)
Mean length (cm)
40
38
36
34
32
30
2012
2014
2016
2018
2020
2022
Year

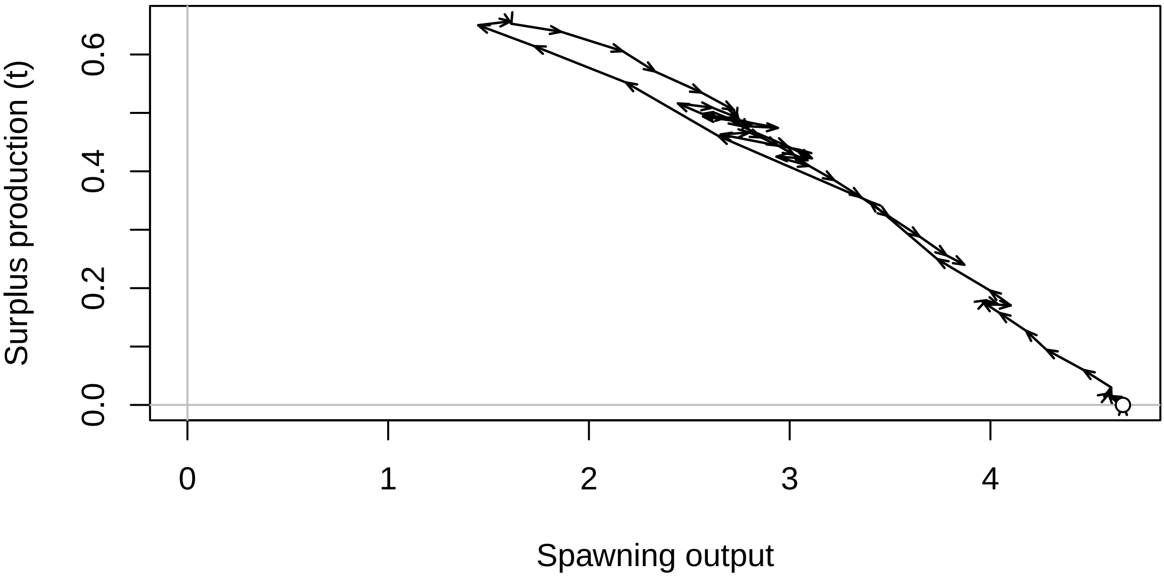


Equilibrium yield (t)
0.0
0.2
0.4
0.6
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of unfished spawning biomass

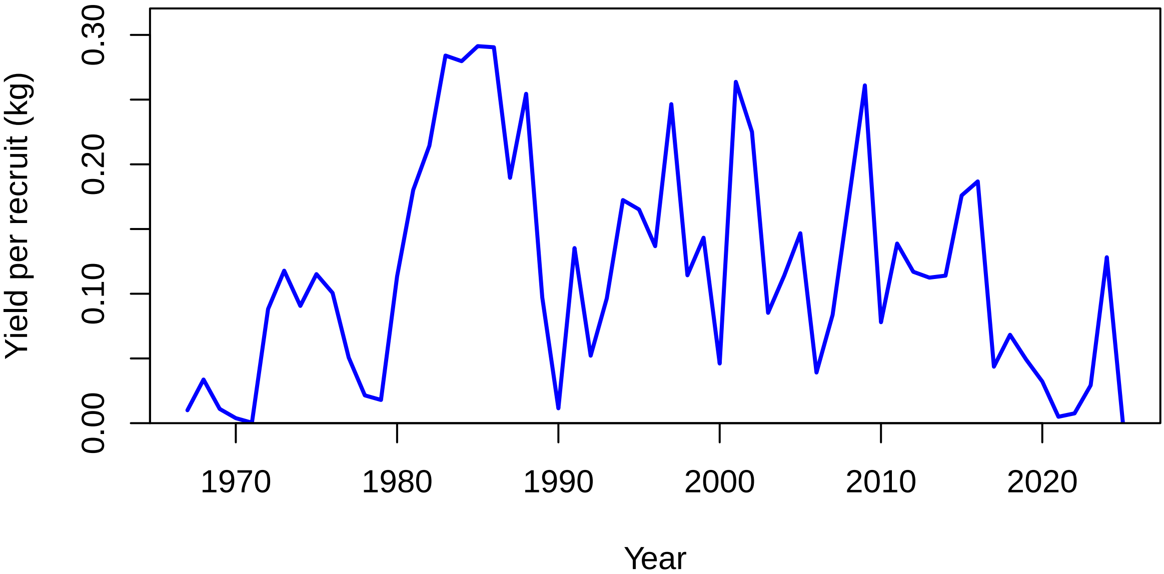

MSY
Btgt
SPR
Current
Equilibrium yield (t)
0.0
0.2
0.4
0.6
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of unfished spawning biomass



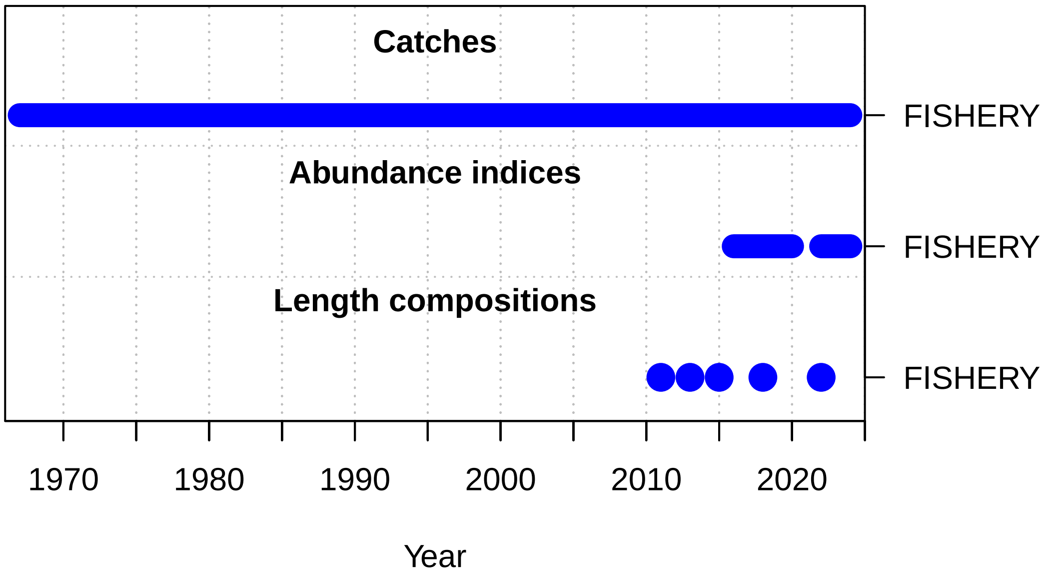
Surplus production (t)
0.0
0.2
0.4
0.6
0
2
4
6
8
10
Total biomass (t)



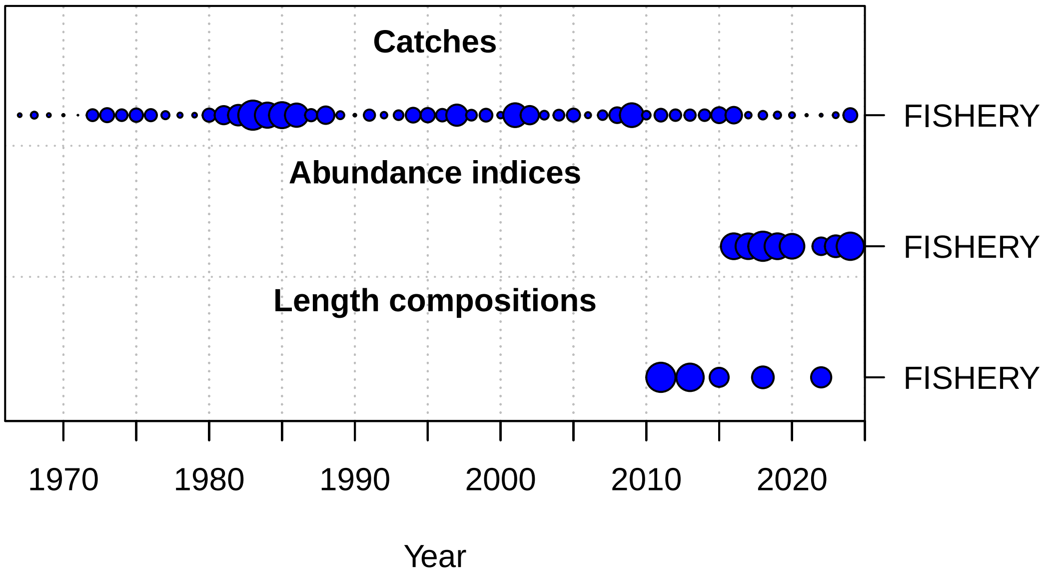
Surplus production (t)
0.0
0.2
0.4
0.6
0
1
2
3
4
Spawning output

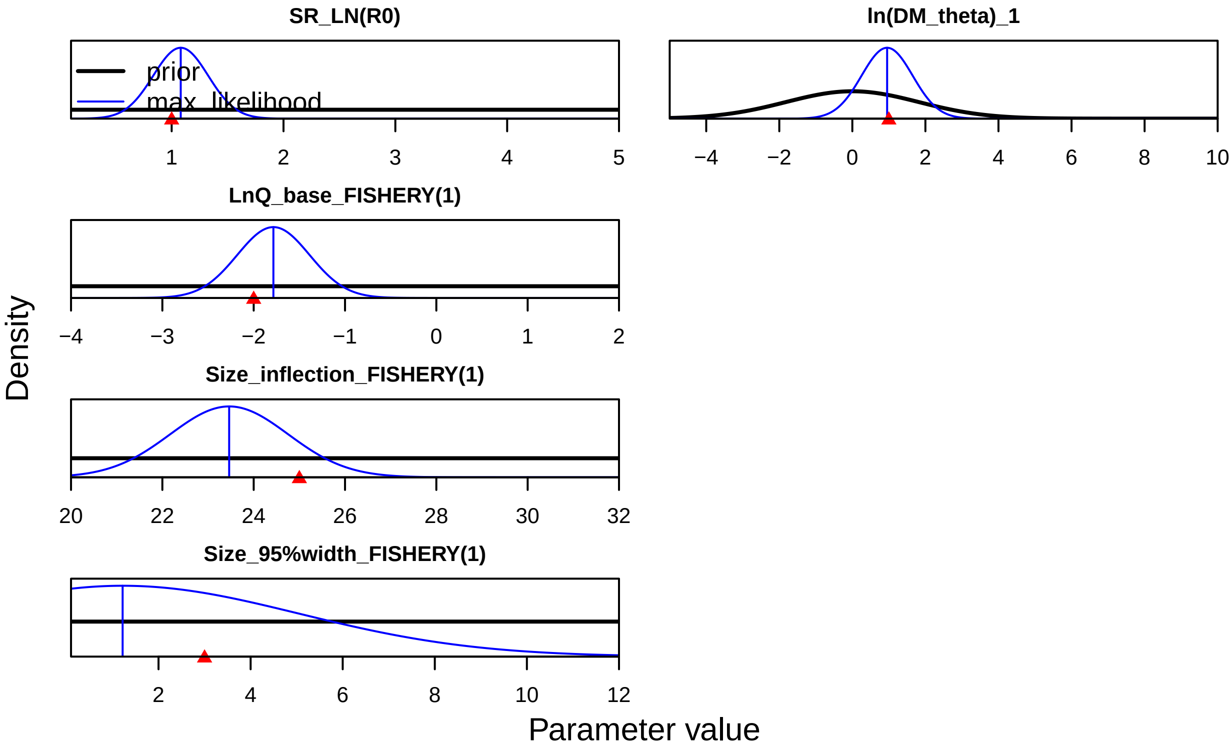


Yield per recruit (kg)
0.00
0.10
0.20
0.30
1970
1980
1990
2000
2010
2020
Year



Catches
FISHERY
Abundance indices
FISHERY
Length compositions
FISHERY
1970
1980
1990
2000
2010
2020
Year


Catches
FISHERY
Abundance indices
FISHERY
Length compositions
FISHERY
1970
1980
1990
2000
2010
2020
Year



SR_LN(R0)
prior
max. likelihood
1
2
3
4
5
ln(DM_theta)_1
−4
−2
0
2
4
6
8
10
LnQ_base_FISHERY(1)
−4
−3
−2
−1
0
1
2
Size_inflection_FISHERY(1)
20
22
24
26
28
30
32
Size_95%width_FISHERY(1)
2
4
6
8
10
12
Density
Parameter value